MILITARY SPECIFICATION

CONTROL AND DISPLAY, TARGETING STATION, M981 (FISTV)

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the interface, performance, and test requirements for the M981 Fire Support Team Vehicle (FISTV) Targeting Station Control and Display (TSCD).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GSS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1260

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SPECIFICATIONS
MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-C-5541	- Chemical Conversion Coating on Aluminum and Aluminum Alloys.
MIL-W-8604	- Welding, Fusion; Aluminum Alloys; Process and Performance of.
MIL-C-22750	- Coating, Epoxy Polyamide.
MIL-P-53022	- Primer, Epoxy Coating, Corrosion Inhibiting, Lead and Chromate Free.
MIL-V-62475	- M981 - FISTV, System Product Fabrication Specification.
MIL-S-62476	- M981 - FISTV, Targeting Station.

ARMY

MIS-34260	- Ground/Vehicular Laser Locator Designator (G/VLLD) P/N 11507293 with P/N 11559640 and FISTV P/N M981 Interface Specification for.

STANDARDS
FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-108	- Basic Requirements for Enclosure of Electrical and Electronic Equipment.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1275	- Characteristics of 28 Vdc Electrical Systems in Military Vehicles.
MIL-STD-45662	- Calibration Systems Requirements.

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MS 3470 - Connectors, Receptacle, Electric Series 2, Crimp Type, Narrow Flange Mount, Bayonet Coupling, Classes A, L, S, and W.

2.1.2 Other Government documents, drawings, and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS
ARMY

12306830 - Targeting Station Control and Display Assembly.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified, the contractor shall furnish samples of the M981-FISTV, Targeting Station Control and Display which shall be subjected to first article inspection (see 4.4 and 6.2). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent targeting station control and diplay units delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Qualified products. The contractor shall be responsible for using parts and assemblies from Qualified Products Lists (QPLs) whenever available Contractor's inspection records shall specifically list all QPL items by number and date of the QPL, name of supplier and part or drawing number(s). When parts and assemblies are approved as qualified products, but not yet listed on the QPL, the contractor shall list the products by number and date of the approved document and name of contractor.

3.3 Materials. Materials shall be as specified in referenced drawings, specifications, and standards. Materials shall be free of defects and imperfections that might affect the serviceability and function of the finished product (see 4.6).

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3.4 Design and construction. The Targeting Station Control and Display (TSCD) unit is designed to function as the data gathering and distribution point, control element, and human interface element for the M981 FISTV Targeting Station as specified in MIL-S-62476. The TSCD interfaces with the North Seeking Gyrocompass (NSG), the Ground/Vehicular Laser Locator Designator (G/VLLD), and the Digital Message Device (DMD). The design of the TSCD front panel shall be as shown in figure 1 (see 4.6.1 through 4.6.3).

3.4.1 Construction. The TSCD shall be constructed in accordance with drawings, parts lists, and other documents listed on Drawing 12306830, together with applicable specifications and standards (see 2.1). If more than one unit of a particular component is used, the units shall be identical in weight, material, and quality. Riveting, welding practices, and quality parameters shall be the same on each unit. Completed TSCDs shall meet all requirements specified herein, and be of a quality that assures a TSCD free of defects that would compromise, limit, or reduce its performance (see 4.6.1 through 4.6.3 and 3.5.4 through 3.5.7).

3.4.2 Major component list. Major components comprising the TSCD assembly are listed below (see 4.6.3).

12306830	Control display assembly - Targeting station.
12306831	Cover assembly - TSCD.
12307097	Circuit card assembly - Display B.
12307099	Circuit card assembly - Display A.
12307104	Control panel assembly.
12307108	Circuit card assembly - Backplane interconnecting board.
12307120	Circuit card assembly - Interface.
12307121	Power supply - TSCD.
12307126	Circuit card assembly - Processor.
12307128	Circuit card assembly - Status.
12307130	Circuit card assembly - Attitude data interface.
12307132	Circuit card assembly - Digital to resolver converter.
12310392	Circuit card assembly - Turret control No. 1.
12311557	Circuit card assembly - Turret control No. 2.
12311621	Wiring harness - TSCD.
12311746	Printed cable assembly - Flexible, control panel.
12311747	Printed cable assembly - Flexible, lower control panel.

3.4.3 Test connector. The TSCD shall be provided with a test connector, designated J3 (P/N MS3470 L22-555), to facilitate diagnostic troubleshooting and fault isolation. Connector J3 pin functions shall be in accordance with figure 2 (see 4.6.3).

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PART NO. 12311715	LEGEND	
-2	CARGO HATCH	
-3	WPN HATCH	LASER INTRPT
-4	DRIVER HATCH	
-5	EL ABOVE	HEAD STOW
-6	EL STOW	AZ STOW
-7	EL BELOW	HEAD ERECT
-8	EQP BAT	VEH BAT
-9		
-10	HEAD READY	NSG ON
-11	NSG FAULT	TSCD FAULT
-12	NSG ALIGN	GLLD FAULT
-13	NSG READY	GLLD ON

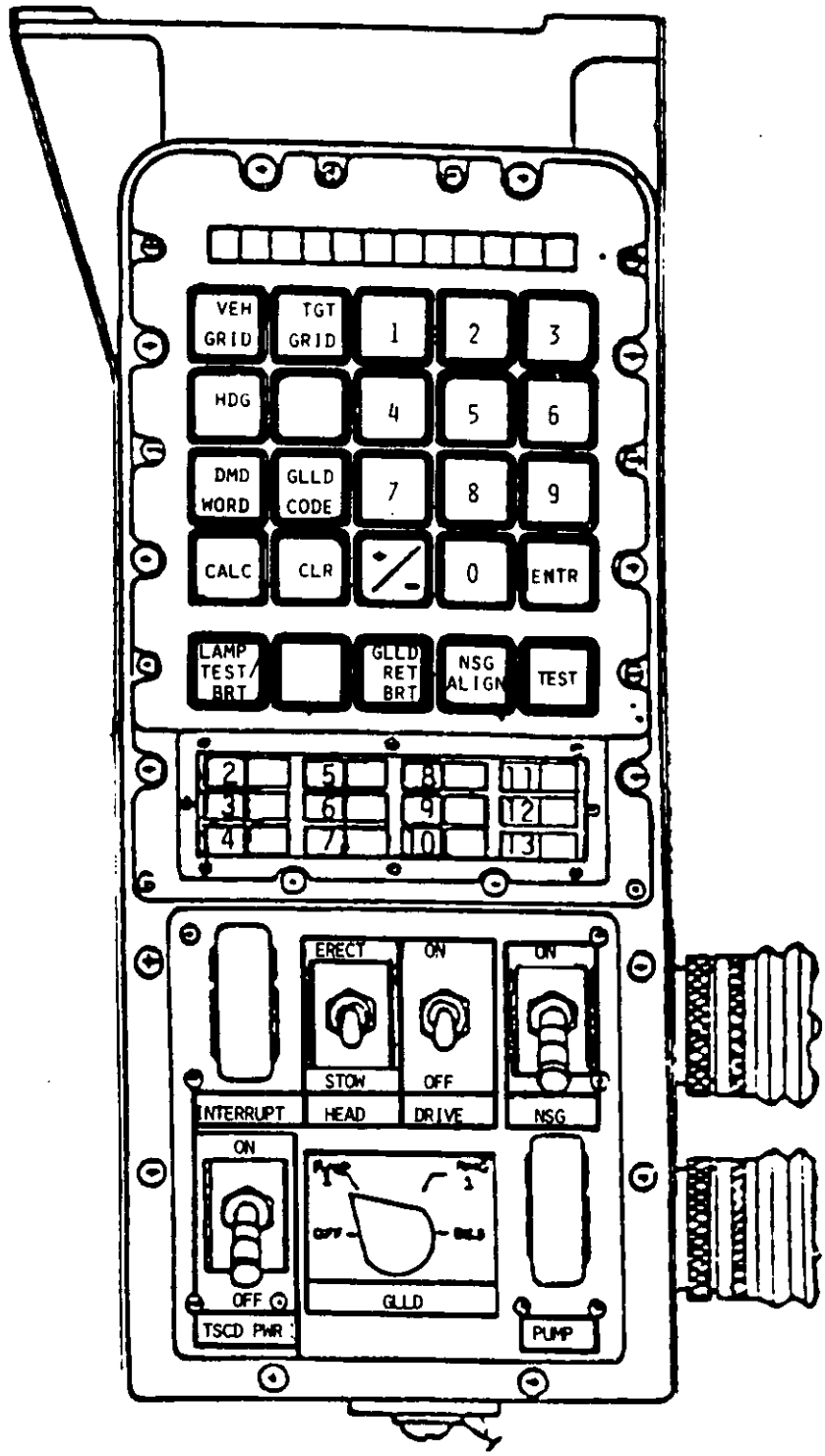


FIGURE 1. TSCD front panel.

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PIN	FUNCTION	PIN	FUNCTION
A	D0	e	MRDB
B	D1	f	SC0
C	D2	g	SC1
D	D3	h	MPQ
E	D4	i	WAIT
F	D5	j	CLEAR
G	D6	k	DMA IN
H	D7	m	DMA OUT
J	MA0	n	Spare
K	MA1	p	MP1RQ
L	MA2	q	ADI DMD
M	MA3	r	ADI GLLD
N	MA4	s	SDO
P	MA5	t	SDI
R	MA6	u	Chassis RTN
S	MA7	v	Spare
T	ADI BSY	w	Spare
U	EF2	x	Spare
V	EF3	y	Spare
W	MPEF4	z	Spare
X	N0	AA	Spare
Y	N1	BB	Spare
Z	N2	CC	+5V lamp
a	2MHZ CLK	DD	+5V logic
b	TPA	EE	+5 RTN
c	TPBB	FF	+12 VDC
d	MWRB	GG	-12 VDC
		HH	+12 RTN

FIGURE 2. Test connector, J3, pin functions.

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3.5 Performance.

3.5.1 Keys, switches, and indicators. The performance of various key/switch functions and indicators, located on the TSCD front panel as shown in figure 1, shall be as specified in the Appendix (see Sections 30 and 40 in Appendix).

3.5.2 Electrical.

3.5.2.1 Voltage. The TSCD shall operate when supplied with 24 ± 6 volts (V) direct current (dc) steady-state voltage. The "EQPT BAT" lamp shall illuminate when the voltage drops below 21 ± 0.5 V dc for greater than 500 millisecond (ms). The "VEH BAT" lamp shall illuminate where the vehicle battery voltage drops below 17.5 ± 0.5 V dc for greater than 500 ms (see 40.1.19 in Appendix).

3.5.2.2 Electromagnetic compatibility. The TSCD shall perform as specified when tested by the method of MIL-STD-462 to the limits given in MIL-STD-461, Notice 4, except as otherwise noted. Applicable requirements are as follows (see 4.8):

CE01	Conducted Emission, Power Leads 30 - 50 kilohertz (kHz)
CE04	Conducted Emission; Power Leads 50 kHz - 50 megahertz (MHz) (broadband noise only)
CE05	Conducted Emissions, Control and Signal Leads 50 kHz - 50 MHz
CE07	Conducted Emissions, Power Leads, Spikes, Time Domain 1.5 - 65 MHz
CS01	Conducted Susceptibility, Power Leads 30 - 50 kHz
CS02	Conducted Susceptibility, Power Leads 50 kHz - 400 MHz
RE02	Broadband Radiated Emissions, 14 kHz - 1 gigahertz (GHz) (levels may exceed MIL-STD-461 limits from 60 kHz - 1 MHz and at 43 MHz)
RS03	Radiated Susceptibility 10 kHz - 12.4 GHz

3.5.3 Reliability. The TSCD shall have a minimum acceptible value (MAV) mission success reliability of 0.9912 where operated in accordance with the composite mission profile as defined in MIL-V-62475 (see 4.6.5).

3.5.4 Environmental conditions. The TSCD, when installed and integrated with the balance of the vehicle system, shall perform as specified when exposed to the following environments (see 4.7).

3.5.4.1 Temperature. The TSCD shall be capable of sustained operation in an ambient temperature range of -25 to +145 degrees Fahrenheit (°F) [-32 to 63 degrees Celsius (°C)]. It shall also be capable of being stored without damage at temperature extremes of -40°F (-40°C) and +160°F (71°C) (see 4.7.1).

3.5.4.2 Shock. The TSCD, when mounted in the vehicle, shall perform as specified when exposed to maximum shocks of 30 gravity units (g) peak in a vertical direction and 15 g's peak in all other directions with a pulse width of 11 ms (see 4.7.2).

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3.5.4.3 Vibration. The TSCD, when mounted in the vehicle and in the operating mode, shall be capable of withstanding vehicle frequency and acceleration inputs (see 4.7.3).

3.5.4.4 Spray tightness. The TSCD shall not be adversely affected when the FISTV in which it is mounted is cleaned by hosing with a high pressure water stream [i.e., 35 pounds per square inch (psi) at 65 gallons per minute (gal/min)] [241.32 kilopascals (kPa) at 246 liters per min (L/min)] in accordance with the spraytight requirements of MIL-STD-108, table II (see 4.7.4).

3.5.4.5 Humidity. The TSCD shall be capable of sustained operation in conditions of 95 percent relative humidity at temperatures in the range of -25 to +145°F (-32 to +63°C) and at humidity conditions of 5 percent relative humidity at temperatures from 70 to 125°F (21 to 52°C) (see 4.7.5).

3.5.4.6 Sand and dust. The TSCD shall not be adversely affected when subjected to windblown sand and dust. Sand particle sizes shall range from 3.93×10^{-4} to 3.93×10^{-2} inch [0.01 to 1.00 millimeter (mm)] in diameter with predominant diameter between 5.91×10^{-3} and 1.18×10^{-2} inch (0.15 and 0.30 mm), with wind velocity greater than 30 knots [56 kilometers per hour (kph)] and the dust particle sizes shall be 3.94×10^{-6} to 3.93×10^{-4} inch (0.0001 to 0.01 mm) at a density of 6×10^{-9} grams per cubic centimeter (cc) with wind velocity of 35 knots (65 kph) (see 4.7.6).

3.5.4.7 Salt fog. The TSCD shall not adversely affected by salt fog in concentrations equivalent to 25 pounds/acre/year (see 4.7.7).

3.5.4.8 Explosive atmosphere. The TSCD, when installed in the vehicle and operating, shall not cause ignition of any ambient-explosive-gaseous mixture of air as might be found to be present (see 4.7.8).

3.5.4.9 Electromagnetic discharge. The TSCD shall be capable of withstanding discharge of personnel-borne electrostatic charges of 100 picofarad capacity, at 15 000 V, discharge through 1500 ohms (see 4.7.9).

3.5.4.10 Pressure. The TSCD shall operate at altitudes from 0 to 10 000 feet [0 to 3.05 kilometers (km)] corresponding to pressures of 14.7 to 10.1 psi (101 to 69.6 kPa). Non-operating, the equipment shall withstand exposure to altitude from 0 to 40 000 feet (0 to 12.2 km) corresponding to pressures of 14.7 to 2.47 psi (101 to 17 kPa) (see 4.7.10).

3.5.4.11 Fungus. The materials used in the TSCD shall be non-nutrients for fungi per MIL-STD-454, requirement 4. If materials which are nutrients for fungi must be utilized, such materials must be approved by the Government (see 4.7.11).

3.5.4.12 Submersion. The TSCD shall remain watertight when submerged in accordance with the spraytight requirements of MIL-STD-108, table III (see 4.7.12).

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3.5.5 Protective finishes. Unless otherwise specified, the exterior surface of the TSCD; except for the front panel, display, and cables, shall be painted light green, No. 24533 per FED-STD-595, with expoxy-polyamide chemical agent resistant coating per MIL-C-22750, type I. All aluminum surfaces shall be cleaned and treated per MIL-C-5541 and primed per MIL-P-53022. Markings shall be black, No. 17038, in accordance with MIL-C-22750 (see 4.5.2.1).

3.5.6 Marking and identification plates. Parts and components of the TSCD shall be marked in accordance with MIL-STD-130. The data plates shall be in accordance with the requirements of MIL-P-514, or applicable drawings (see 4.5.2.1).

3.5.7 Workmanship. Workmanship shall be of a quality to assure that the TSCD and its components are free of defects that compromise, limit, or degrade performance in its intended use. Electrical wires and hydruaiic lines shall be located in a manner as to prevent possible damage by rubbing on adjacent lines or appendages. No workmanship condition shall exist which may create a safety hazard to operating or maintenance personnel. Particular attention shall be paid to the marking of parts, plating, machine-screw assemblage, welding, brazing, and freedom of parts from burrs and sharp edges (see 4.5.2.1).

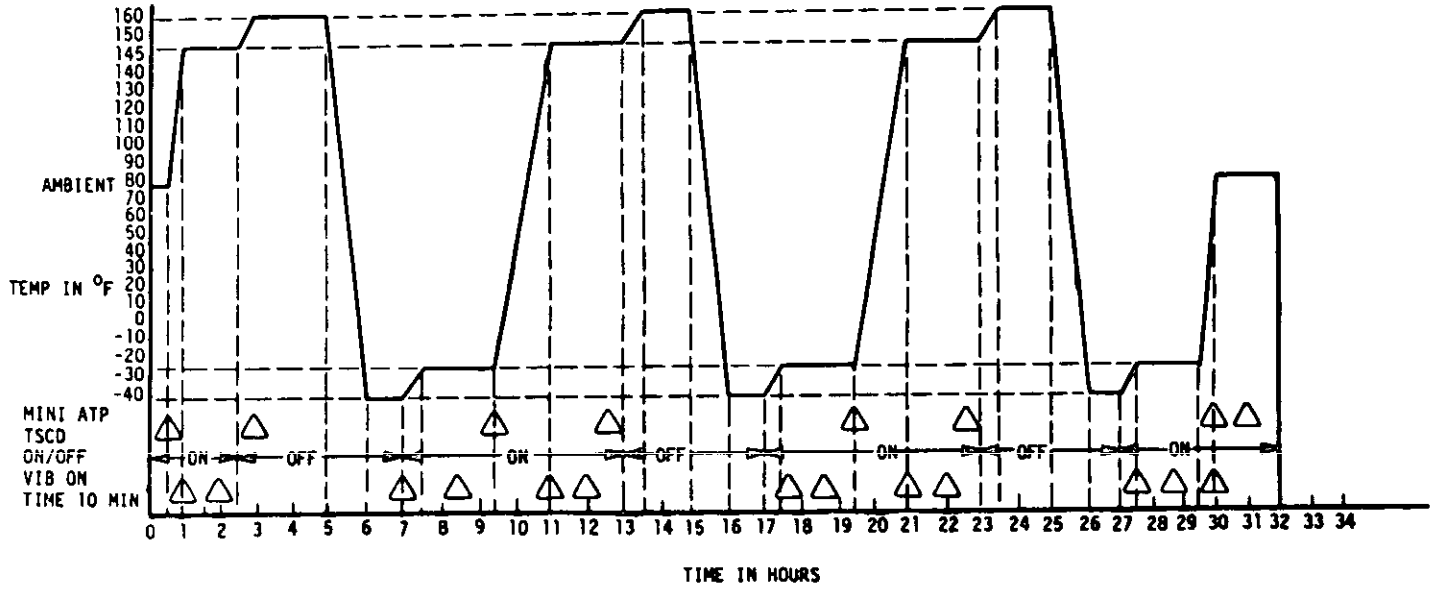
3.5.8 Welding repairs. Welding repairs of any type or class shall be made only when, and to the extent, specifically authorized by the applicable drawings, specifications, or welding procedures which have been approved by the procuring activity. Welding of aluminum alloys shall be in accordance with MIL-W-8604 (see 4.6.7).

3.5.9 Preconditioning. Each production unit shall be preconditioned to ensure proper workmanship and to preclude infant mortality failures. Preconditioning shall consist of combined temperature cycling and vibration testing per figure 3. Thirty-two (32) hours shall be completed followed by a shock test and a leakage test. Bolts used to mount the TSCD to the test fixture shall be torqued to 70 pounds force inch (lbf in) [7.91 Newton meters (N m)]. A vibration cycle shall consist of 2.2 g peak acceleration at a frequency of 60 hertz (Hz). Vibration shall be applied along the vertical axis as indicated in figure 3 for 10 minute periods. The TSCD shall be subjected to a shock test in accordance with MIL-STD-810, method 516.2, procedure 1. Three half-sine shocks in each direction for each axis (18 shocks total) are required. The peak value shall be 15 g's with a duration of 11 ms. The leakage test shall be conducted in accordance with MIL-STD-108, table III, spray tight. Any evidence of water inside the TSCD shall constitute a failure.

3.5.10 Maintainability. The TSCD shall be capable of meeting the following requirements (see 4.9).

3.5.10.1 Maximum time to repair (MTTR). The maximum time to return the TSCD to an operational condition at an organizational level maintenance shall be 0.5 hours, with 90 percent confidence.

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NOTE:
1. Mini ATP indicates "Mini Acceptance Test Procedure".

FIGURE 3. TSCD Preconditioning.

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3.5.10.2 Maintenance ratio (MR). The TSCD shall have a MR [maintenance manhours (MMH) divided by operating hours] that does not exceed 0.03 for organizational level maintenance and 0.06 for direct support/general support (DS/GS) level maintenance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of surveillance of the operation to determine that practices, methods, and procedures of the written inspection plan are being properly applied; and Government product inspection to determine quality of the product offered for acceptance. Deviation from the prescribed or agreed upon procedures, or instances of poor practices which might have an affect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to specified criteria has been demonstrated.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
   1. Preproduction inspection (see 4.4.1).
   2. Initial production inspection (see 4.4.2).

b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Acceptance tests (see 4.5.3.1).

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4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 23 ± 10°C
b. Barometric pressure 725 + 50mm Hg
- 75
c. Relative humidity 50 ± 30 percent

4.4 First article inspection. First article inspections shall be performed on preproduction or initial production samples as specified herein. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply targeting stations that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 Preproduction inspection. When specified (see 6.2), the TSCD produced under a contract shall undergo a preproduction inspection prior to full scale production. The sample targeting station shall be subjected to the examinations and tests specified in 4.5.2.1 and table I.

4.4.2 Initial production inspection. Two TSCDs from the first 20 produced, shall be subjected to the initial production inspection as specified in 4.5.2.1 and table I.

4.4.3 TSCD disposition. On completion of the preproduction inspection, the TSCD unit shall remain at the manufacturing facility as a production sample and be the last unit shipped under the contract. The unit may be released sooner at the discretion of the contracting officer. The contractor shall service and maintain the TSCD during this period in accordance with applicable documents for care and preservation while in storage.

4.4.4 First article inspection failure. Test item deficiencies during, or as a result of, the first article test, shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the first article test, shall be prima facie evidence that all items already produced prior to completion of the first article test are similarly deficient unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such deficiencies on all items shall be corrected by the contractor at no cost to the Government. The Government shall not final accept products until first article testing is completed to the satisfaction of the Government.

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the TSCD units of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

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4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105. Before sampling may be initiated, the contractor shall establish by examination of at least 20 consecutively produced targeting stations that the process average percent defective, as defined in MIL-STD-105, is not greater than the specified AQLs.

4.5.2 Quality conformance examinations.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) on the basis of percent defective for the various defects listed in table II.

Classification	AQL
Major	25/100 units
Minor	150/100 units

TABLE I. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance	
				Examination	Acceptance
Initial production units	3.1	4.4.2	X		
Materials	3.3	4.5.2.1		X	
Design and construction	3.4	4.5.2.1		X	
Production drawings	3.4.1	4.5.2.1		X	
Major component list	3.4.2	4.6.3			X
Test connector	3.4.3	4.5.2.1		X	
Voltage	3.5.2.1	40.1.19	X		X
Electromagnetic interference	3.5.2.2	4.8	X		
Reliability	3.5.3	4.6.5	X		
Temperature	3.5.4.1	4.7.1	X		
Shock	3.5.4.2	4.7.2	X		
Vibration	3.5.4.3	4.7.3	X		
Spray tightness	3.5.4.4	4.7.4	X		
Humidity	3.5.4.5	4.7.5	X		
Sand and dust	3.5.4.6	4.7.6	X		
Salt fog	3.5.4.7	4.7.7	X		
Explosive atmosphere	3.5.4.8	4.7.8	X		
Electrostatic discharge	3.5.4.9	4.7.9	X		
Pressure	3.5.4.10	4.7.10	X		
Fungus	3.5.4.11	4.7.11	X		
Submersion	3.5.4.12	4.7.12	X		

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TABLE I. Classification of inspections - Continued.

Title	Requirement	Inspection	First article	Quality conformance	
				Examination	Acceptance
Protective finishes	3.5.5	4.5.2.1		X	
Markings and data plates	3.5.6	4.5.2.1		X	
Workmanship	3.5.7	4.5.2.1		X	
Maintainability	3.5.10	4.9	X		
Packaging	5.1	6.2	X		X
Display	30.3	40.1.2	X		X
ENTR key	30.5	40.1.3	X		X
CLR key	30.6	40.1.4	X		X
VEH GRID key	30.7.1	40.1.5	X		X
TGT GRID key	30.7.2	40.1.6	X		X
DMD Word key	30.7.3	40.1.8	X		X
CALC key	30.7.4	40.1.10	X		X
HDG key	30.7.5	40.1.7	X		X
GLLD CODE key	30.7.6	40.1.9	X		X
LAMP TEST/BRT key	30.7.7	40.1.11	X		X
GLLD RET BRT key	30.7.8	40.1.12	X		X
NSG ALIGN key	30.7.9	40.1.13	X		X
TEST key	30.7.10	40.1.4	X		X
Control of GLLD function	30.8	40.1.15	X		X
Built in test (BIT)	30.9	40.1.16	X		X
Operator error messages	30.10	4.5.2.1	X		X
Indicators	30.11	40.1.17	X		X
Interrupts	30.11.3	40.1.18	X		X
Control switches	30.12	4.5.2.1	X		X

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as shown in table II.

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TABLE II. Classification of defects.

Category	Defect	Method of examination
Major	AQL: 25 Defects/100 Units	
101	Configuration not as specified.	Visual
102	Parts or components missing or not as specified.	Visual
103	Materials not as specified.	Visual
104	Safety devices missing or not as specified.	Visual and functional
105	System interfaces not as specified.	Functional
106	Treatment and painting not as specified.	Visual
107	Workmanship not as specified.	Visual
108	Assembly incomplete or incorrect	Visual
109	Identification marking missing, incomplete, or not legible.	Visual
Minor	AQL: 150 Defects/100 Units	
201	TSCD; improper assembly or installation.	Visual
202	Fasteners; improper assembly, torque.	SIE 1/
203	Wiring; defective, missing, improper installation, coding or protection.	Visual and functional
204	Decals, marking data, and instruction plates; missing, improper location or size, defective.	Visual
205	Paint; application or color improper.	Visual

1/ SIE = Standard Inspection Equipment.

4.5.2.3 Examination failure. If a TSCD fails to pass any examination specified herein, the Government will stop acceptance until evidence has been provided by the contractor that corrective action has been taken.

4.5.2.4 Recurring major deficiencies. (Applicable only when inspection is performed on a sampling basis.) A major deficiency is recurring when the same defect occurs more than once in the same sample or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or approved equivalent) reflects such a condition. Recurring major deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. The deficiencies shall be corrected when found.

4.5.2.5 Recurring minor deficiencies. (Applicable only when inspection is performed on a sampling basis.) A minor deficiency is recurring if it occurs more than twice in the same sample or when the defect occurs in four successive samples. Recurring minor deficiencies shall be cause for the entire lot or lots to be inspected for recurring deficiencies. Correction of all deficiencies shall be accomplished prior to acceptance.

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4.5.3 Quality conformance tests.

4.5.3.1 Acceptance tests (100 percent). To verify conformance to Section 3, each TSCD shall be subjected to the acceptance tests as specified in table I.

4.5.3.2 Failure. Failure of any TSCDs to pass any of the specified inspections shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.6 Methods of inspection.

4.6.1 Materials and construction. Conformance to 3.3 through 3.4.3, and 3.5.4 through 3.5.7, shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.6.2 Defects. Conformance to 3.3 through 3.4.3, and 3.5.5 through 3.5.7, shall be verified by examination for the defects listed in table II. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.6.3 Parts and components. Parts, components, and assemblies shall be inspected for conformance to requirements of drawings subordinate to 12306830. When applicable, inspection shall be in accordance with Supplementary Quality Assurance Provisions (SQAP's) or Quality Assurance Requirements (QAR's) (see 6.2). Inspection shall be performed during the appropriate stages of manufacture.

4.6.4 Performance. Conformance to 3.5 through 3.5.4.12, shall be verified by effecting the various operational test modes and testing procedures specified in Sections 4.7 and 4.8.

4.6.5 Reliability verification. To determine conformance to 3.5.3, a point estimate value equal to or greater than the value specified by applicable reliability data shall be demonstrated during initial production inspection in 4.4.3. The point estimate will be computed by dividing the total cummulative test hours of all test TSCDs by the total cummulative chargeable failures experienced on these TSCDs.

4.6.6 Welding repairs inspection. To verify conformance to 3.5.8, a visual inspection of welding repairs made shall be performed. Applicable records of all inspection results shall be maintained, together with dispositions made of repaired parts.

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4.7 Environmental test (see 3.5.4).

4.7.1 Temperature tests. To verify conformance to 3.5.4.1, the TSCD shall be subjected to a low temperature test in accordance with MIL-STD-810, method 502, procedure 1. Storage temperature shall be -40°F (-40°C) for a duration of 6 hours. After storage, the TSCD temperature shall be raised to -25°F (-32°C), stabilized and operated. Failure of the TSCD to meet the specified operational requirements shall constitute failure of this test. The TSCD shall also be subjected to a high temperature test in accordance with MIL-STD-810, method 501, procedure 1. Storage temperature shall be 160°F (71°C) for a duration of 4 hours. After storage, the equipment shall be lowered to 145°F (63°C), stabilized and operated. Failure of the TSCD to meet the specified operational requirements shall constitute failure of this test.

4.7.2 Shock test. To verify conformance to 3.5.4.2, the TSCD shall be subjected to a shock test in accordance with the requirements of MIL-STD-810, method 516.2, procedure 1, for ground equipment except that the maximum g levels shall be 15 g for all but the vertical direction which shall be 30 g.

4.7.3 Vibration test. To verify conformance to 3.5.4.3, the TSCD shall be subjected to a vibration test in accordance with the requirement of MIL-STD-810, procedure VIII, curve V of figure 514.2-6 except the highest frequency shall be 500 Hz. Sinusoidal cycling time shall be 3 hours per axis. The TSCD shall withstand exposure to the following random inputs with the maximum exposure time of 16.8 hours per axis.

Frequency (Hz)	Power Spectral Density (g^2/Hz)
75 to 150	0.0027
150 to 300	0.00137
300 to 1000	0.00092

4.7.4 Spray tightness test. To verify conformance to 3.5.4.4, the TSCD shall be subjected to a water spray test in accordance with MIL-STD-108, table II. The TSCD shall be operated after the test and display no degradation of performance.

4.7.5 Humidity test. To verify conformance to 3.5.4.5, a humidity test shall be performed in accordance with MIL-STD-810, method 507.1, procedure III, with the exception that the high temperature shall be 145°F (63°C) and the low temperature shall be -25°F (-32°C). The test shall be repeated with a low humidity of 5 percent at temperatures from +70 to +125°F (21 to 52°C). Steps 7 through 9 of procedure III are not required for either test.

4.7.6 Sand and dust test. To verify conformance to 3.5.4.6, a sand and dust test shall be performed in accordance with MIL-STD-810, method 510, procedure I, with the exception that sand and dust particle sizes shall be as specified in 3.5.4.6.

4.7.7 Salt fog test. To verify conformance to 3.5.4.7, a salt fog test shall be performed in accordance with MIL-STD-810, method 509, procedure I.

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4.7.8 Explosive atmosphere test. To verify conformance to 3.5.4.8, the TSCD shall be subjected to an explosive atmosphere test in accordance with MIL-STD-810, method 511.1, procedure I. Test shall be performed at 5000, 10 000, and 15 000 feet (1.5, 3.1, and 4.6 km), and a temperature of 158 ± 6°F (70 ± 3°C). The fuel mixture shall be injected, when chamber walls are within 30°F (11°C) of chamber ambient, for a period of 3 minutes. This test may be deleted if the TSCD is found to be properly sealed as a result of the spray tight test (see 4.9.4).

4.7.9 Electrostatic discharge test. To verify conformance to 3.5.4.9, the TSCD shall be subjected to an electrostatic discharge of 100 picofarads capacity at 15 000 V, discharged through 1500 ohms.

4.7.10 Pressure test. To verify conformance to 3.5.4.10, the TSCD shall be subjected to a pressure test in accordance with MIL-STD-810, method 500.1, procedure I, with the exception that the maximum altitude while operating shall be 10 000 feet (3.05 km).

4.7.11 Fungus test. To verify conformance to 3.5.4.11, the TSCD shall be tested at 84°F (29°C) and 95 percent relative humidity as specified in MIL-STD-810, method 508.1, procedure I, for a period of 90 days. This test may be waived if certification is provided that only non-nutrient materials per MIL-STD-454, requirement 4, have been utilized.

4.7.12 Submersion test. To verify conformance to 3.5.4.12, the TSCD shall be subjected to a spray tight test in accordance with MIL-STD-108, table III. Drain holes shall be temporarily sealed for the water submergence test. The TSCD shall be operated and display no degradation of performance after the test.

4.8 Electromagnetic interference test. To verify conformance to 3.5.2.2, the TSCD shall be subjected to the applicable tests specified in MIL-STD-462.

4.9 Maintainability. To determine conformance to 3.5.10, a maintenance ration (MR) equal to or less than specified in 3.5.10.2 must be demonstrated during initial production testing in 4.4.2. The MR will be based upon the total cummulative maintenance manhours with the exception of human induced errors and driver/crew checks and services.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking of the TSCD units for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

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6. NOTES

6.1 Intended use. The TSCD units covered by this specification are intended for use by military ground forces as a major component of the M981 FISTV. The TSCD shall be used for target data gathering, data processing, and distributing grid direction of the target to other fighting units for fire control purposes. It shall also provide diagnostic capability for the internal processor system as well as for other interfacing systems.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, drawing number of required item.
c. If first article samples are not required (see 3.1).
d. If responsibility for inspection shall be other than as specified (see 4.1).
e. If responsibility for inspection equipment shall be other than as specified (see 4.1.1).
f. If inspection conditions shall be other than as specified (see 4.3).
g. If preproduction inspection is required (see 4.4.1).
h. Selection of applicable level and packaging standard or packaging data (see 5.1).
i. If painting, marking, and data plates are to be other than specified (see 3.5.6).

6.3 Supersession data. This military specification supersedes Purchase Description M981PD-C-62479.

Custodian:
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Preparing activity:
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Performance Requirements - Key/Switch Functions and Indicators
Targeting Station Control and Display (TSCD)

10. SCOPE

10.1 Scope. This appendix specifies performance requirements for the various key/switch functions and indicators provided on the TSCD front panel as shown in figure 1 of this specification. It provides menus and sequences for tests to be performed to verify operational capability of the TSCD unit to meet specified requirements. Test sequences are included to determine integrated performance of the TSCD unit with other operational components of the M981 Fire Support Team Vehicle (FISTV) Targeting Station system. This appendix shall be construed to be a mandatory part of this specification.

20. REFERENCED DOCUMENTS

20.1 Referenced documents. This section is not applicable to this appendix.

30. REQUIREMENTS

30.1 Requirements. The requirements cited in this appendix shall be supplemental to those included under Section 3.5 of this specification entitled Performance.

30.2 Mandatory status. The requirements of this appendix shall be mandatory together with all others included in this specification.

30.3 Display. The front panel display shall provide a single row of up to 12 alphanumeric characters. The display shall show angular values specified in military mils [least significant digit (LSD) = 1 mil) and distances/position data in meters (LSD = 1 meter, except as noted elsewhere). Alphanumeric characters shall prompt the TSCD operator during the performance of TSCD related functions.

30.4 Numeric entry items. Menu items allowing numeric entry shall consist of a text legend on the left of the display and a numeric field on the right. The length of the numeric field for a particular item shall be the length (including sign) of the largest allowable number for that particular item. Displayed numbers shall be right justified in the numeric field with leading zeros suppressed. Negative numbers shall be preceded by a minus (-) sign, while positive numbers shall have no sign. A numeric entry item can be changed by entering a sequence of numeric keys terminated by the ENTR key. Upon depressing the first numeric key, the numeric field shall be cleared and the first digit (or sign) shall be entered at the right of the

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display. The numeric field is now open and only numeric keys, CLR, or ENTR shall be recognized. Succeeding digits are entered at the right of the display as the preceding digits are shifted left. When the numeric field has been filled, either a CLR or ENTR key must be used (see 30.5 and 30.6). When the numeric digit entry exceeds the preprogrammed upper or lower field limits, an operator error message shall be immediately displayed (see table III).

30.4.1 Terminating numeric entry. The numeric field shall be closed by pressing either CLR or ENTR. When ENTR key is pressed, the new number shall be checked against the preprogrammed upper and lower limits, and if valid, shall be stored in memory in place of the previously stored data. If the entry is not valid, an operator error message shall be displayed (see table III). If the CLR key is depressed, the new number shall be discarded and the previous number shall be redisplayed.

TABLE III. TSCD operator error messages.

Display	Cause
? E>878 KM	Easting value entered was greater than 878 km.
? E<122 KM	Easting value entered was less than 122 km.
N OUT OF RANGE	Northing value entered was greater than 9 999 999 in absolute value.
? ALT > 7500 M	Altitude value entered was greater than 7500 meters.
? ALT < -400 M	Altitude value entered was less than -400 meters.
AZ > 6399 MIL	Azimuth value entered was greater than 6399 mils.
AZ < 0 MIL	Azimuth value entered was negative.
RNG < 400 M	Range value entered was less than 400 meters.
OUT OF RANGE	Elevation value entered is greater than 999 mils.
? GLLD CODE	GLLD code number entered contained a 0 or 9, or had less than 3 digits.
NO VALID E/N	An update, realign, or reinitialize NSG command was attempted before vehicle northings and eastings were entered.
NSG NOT RDY	A polar data request or a realign NSG command was attempted before the NSG was in the Directional-Gyro mode.
TOO MANY	Number entered had too many digits.
TRIG SHORT	Trigger switch in hand control is activated (depressed or electrically shorted) during power-up.

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TABLE III. TSCD operator error messages - Continued.

Display	Cause
? V EASTINGS	No vehicle easting value entered.
? V NORTHINGS	No vehicle northing value entered.
? V ALTITUDE	No vehicle altitude value entered.
? T EASTINGS	No target eastings value entered.
? T NORTHINGS	No target northing value entered.
? T ALTITUDE	No target altitude value entered.
NO VALD TE/N	No valid target eastings/northing data.
NO VALD VE/N	No valid vehicle eastings/northing data.

30.5 ENTR (enter) key. The ENTR key shall perform the following functions:

a. For numeric entry items, pressing ENTR shall terminate the entry of a numeric string per 30.4.1. After entering a valid number, the next item in the menu shall be displayed, unless the number was entered into the last menu item, in which case "KEYBOARD RDY" shall be displayed. If the numeric field has not been re-opened, the number present on the display shall be entered unmodified.

b. For other than numeric entry items, pressing ENTR shall perform the action as specified in the individual function key descriptions.

c. If an operator error message is displayed, pressing ENTR shall have no effect.

30.6 CLR (clear) key. The CLR key shall provide the following functions:

a. Shall clear an operator error message and then revert back to displaying the previous unmodified menu item.

b. If a numeric field is open, the field shall be closed, the new number discarded, and the previous number shall be redisplayed.

c. If neither of the above, the CLR key shall cause the TSCD to display "KEYBOARD RDY".

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30.7 Function keys. The function keys are identified as VEH GRID, TGT GRID, DMD WORD, CALC, HDG, GLLD CODE, LAMP TEST/BRT, GLLD RET BRT, NSG ALIGN, and TEST; keyboard layout and arrangement shall be as shown in figure 1. Each function key is associated with a menu of one or more items. Depressing a function key for a menu different than that which is displayed shall cause the first item of the new menu to be displayed. Repeatedly depressing the same function key shall cause each successive item of that menu to be displayed until the last item, after which "KEYBOARD RDY" shall be displayed, followed by the first item, and so on. Except for the HDG key, no modification of data or any other action shall occur while stepping through the menu in this manner.

30.7.1 VEH GRID (vehicle grid) key. The VEH GRID key shall provide a menu for entering and examining Universal Transverse Mercator (UTM) coordinates of the FISTV. These coordinates shall be used in the calculation routines (see 30.7.4) and by the NSG as specified below. Power-up values shall be as follows:

Vehicle Eastings/Northings	########
Vehicle Eastings	######
Vehicle Northings	########
Vehicle Altitude	####

Menu description

Item 1 V EN e e e e n n n n
Numeric entry: NO
Four Eastings digits (eeee) and four Northings digits (nnnn), to the nearest ten meters, for quick reference. The sign of the Northings value is not shown. Pressing ENTR shall have no effect.

Item 2 V E e e e e e e
Numeric entry: YES; Field Length (FL) = 6
Entry and display of vehicle Eastings to the nearest meter.

Item 3 V N (-) n n n n n n n
Numeric entry: YES; FL = 8
Entry and display of vehicle Northings to the nearest meter.
[NOTE: Southern hemisphere northings are preceeded by a minus (-).]

Item 4 V A (-) a a a a
Numeric entry: YES; FL = 5
Entry and display of vehicle Altitude (from sea level) in meters. A negative value indicates below sea level. Upon entering a valid altitude, and if valid vehicle Eastings and Northings have been entered, an update command shall be sent to the NSG.

30.7.2 TGT GRID (target grid) key. The TGT GRID key shall provide a menu for entering and examining UTM coordinates of a target. These coordinates shall be used in the calculation routines (see 30.7.4). Power-up values shall be as follows:

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Target Eastings/Northings	########
Target Eastings	######
Target Northings	########
Target Altitude	####

Menu description

Item 1 V EN e e e e n n n n
Numeric entry: NO
This item provides four Eastings digits (eeee) and four Northings digits (nnnn), to the nearest ten meters, for quick reference. The sign of the Northings value is not shown. Depressing ENTR shall not have an effect.

Item 2 V E e e e e e e
Numeric entry: YES; FL = 6
This item allows entry and display of target Eastings to the nearest meter.

Item 3 V N (-) n n n n n n n
Numeric entry: YES; FL = 8
Entry and display of target Northings to the nearest meter.
[NOTE: Southern hemisphere northings are preceeded by a minus (-).]

Item 4 V A (-) a a a a
Numeric entry: YES; FL = 5
Entry and display of target Altitude (from sea level) in meters. A negative value indicates below sea level.

30.7.3 DMD WORD key. The DMD WORD key shall provide a menu for examining and modifying the DMD polar data word (azimuth, range, and elevation), and for sending the displayed data to the DMD. There are three ways in which the DMD WORD data may be modified:

a. Azimuth, range, and elevation shall be updated when the G/VLLD is fired in a Range mode. The same data is sent to the DMD.

b. Azimuth and elevation may be updated by depressing ENTR while in the Heading mode (see 30.7.5).

c. Azimuth, range, and elevation data may be modified via keyboard entry by the operator using this menu. Power-up values shall be as follows:

Azimuth	0 mils
Range	999 meters
Elevation	0 mils

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Menu description

Item 1 a a a a r r r (-) v v v
Numeric entry: NO
a a a a = azimuth in mils
r r r = range in tens of meters
(-) v v v = vertical angle (elevation) in mils
Current DMD polar data word is displayed. The data displayed shall not be automatically updated when the G/VLLD is fired in the range mode. Pressing ENTR shall have no effect.
[NOTE: Elevation below zero, with respect to the vehicle deck, is preceeded by a minus (-).]

Item 2 DIR a a a a
Numeric entry: YES; FL = 4
Entry and display of DMD word azimuth data. Least significant digit (LSD) = 1 mil. The displayed value shall not be updated when the G/VLLD is fired in a range mode.

Item 3 S L T D I S T r r r r
Numeric entry: YES; FL = 3
Entry and display of DMD range data. LSD = 10 meters. The displayed value shall be updated when the G/VLLD is fired in a range mode.

Item 4 V A (-) v v v
Numeric entry: YES; FL = 4
Entry and display of DMD word elevation (vertical angle) data. LSD = 1 mil. The displayed value shall not be updated when the G/VLLD is fired in the range mode.

Item 5 S E N D (E N T E R)
Numeric entry: NO
Pressing ENTR shall send the current DMD word to the DMD.

30.7.4 CALC (calculate) key. The CALC key shall provide a menu to perform the following calculations:

a. Calculate target position (UTM coordinates) giving vehicle position in UTM coordinates and the DMD polar data word.

b. Calculate FIST vehicle position in UTM coordinates giving a target position in UTM coordinates and DMD polar data word for that target.

Menu description

Item 1 C V L A S T T G T
Numeric entry: NO
Pressing ENTR shall calculate the vehicle position as specified above. The result shall be stored in the VEH GRID menu, and the second item of this menu shall be displayed upon completion of the calculation.

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Item 2 C T L A S T T G T
Numeric entry: NO
Depressing ENTR shall calculate the target position as sepcified above. The result shall be stored in the TGT GRID menu, and the first item of this menu shall be displayed upon completion of the calculation.

30.7.5 HDG (heading) key. The HDG key shall provide continuous azimuth and vertical angle readouts, at an update rate of 4 times per second, as described in the following paragraphs:

a. When the targeting head is erected, the display shall read:

H T a a a a (-) v v v

where "a a a a" is the azimuth angle in mils, "(-) v v v" is the vertical angle in mils. Depressing ENTR while in this mode shall enter the displayed data into the DMD WORD azimuth and vertical angle memory.

b. When the targeting head is stowed, the display shall read:

H V a a a a

where "a a a a" is the vehicle heading in mils. Depressing ENTR shall have no effect.

30.7.6 GLLD CODE key. The GLLD CODE key shall provide a menu to enter and examine the three-digit GLLD CODE. The GLLD CODE interface shall be in accordance with MIS-34260. The power-up value shall be ###.

Menu description

Item 1 G L L D C O D E g g g
Numeric entry: YES; FL = 3
Entry and display of the three-digit GLLD CODE (ggg). Three digits must be entered with the range of each digit being from 1 to 8, inclusive.

30.7.7 LAMP TEST/BRT (lamp test/brightness) key. The LAMP TEST/BRT key shall provide a menu to examine and change the lamp brightness value, and to perform a lamp test of all display and panel lamps. The power-up value for lamp brightness is 9 (full brightness).

Menu description

Item 1 L A M P B R T 9
Numeric entry: YES; FL = 1
Entry and display of the lamp brightness value. Maximum brightness shall be 9 and minimum brightness shall be 0.

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Item 2 L A M P T E S T
Numeric entry: NO
Depressing ENTR shall initiate a lamp test. The alphanumeric display shall be tested by lighting each dot in each character in three groups of four characters, at a rate of one goup per second. After testing the display, all panel lamps shall illuminate for three seconds.

30.7.8 GLLD RET BRT (GLLD reticle brightness) key. The GLLD RET BRT key shall provide a menu to enter and examine the GLLD reticle brightness value. A brightness value of zero shall set the reticle brightness to the same level as a value of one, but shall cause the digital display in the GLLD optics to be turned off except when the laser is fired. Maximum brightness shall be 9. The power-up brightness value shall be one.

Menu description

Item 1 G L L D R E T 1
Numeric entry: YES; FL = 1
Entry and display of the GLLD reticle brightness.

30.7.9 NSG ALIGN key. The NSG ALIGN key shall provide a menu for realigning and reinitializing the north seeking gyrocompass (NSG).

Menu description

Item 1 N S G R E - A L I G N
Numeric entry: NO
Depressing ENTR shall send a realign command to the NSG. An error message shall be displayed, and no realign command sent, if either no vehicle position data has yet been entered, or if the NSG is not in the Directional Gyro (DG) mode.

Item 2 N S G R E - I N I T
Numeric entry: NO
Depressing ENTR shall send a reinitialize command to the NSG. An error message shall be displayed, and no reinitialize command sent, if no vehicle position data has yet been entered.

30.7.10 TEST key. The TEST key shall provide a menu to initiate the "SYSTEM-TEST", examine the current and accumulated built-in-test (BIT) status error codes, clear any existing fault indications, and observe the NSG mode and self-test status. The TSCD shall display BIT status by means of three error codes; a TSCD error code, a GLLD error code, and a NSG error code. An error code is a two-digit hex number which shall be interpreted as follows:

In the binary representation of the error code, each bit shall represent a specific test. A set (1) bit means the test failed, while a reset (0) bit means the test passed. In this manner, up to eight tests can be represented with one error

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code. Failures of more than one test shall be indicated, but not the number of times that a particular test has failed. The particular tests for each error code shall be numbered 0 [least significant (LS) bit] through 7 [most significant (MS) bit], which are listed in table IV.

Menu description

Item 1 S Y S T E M T E S T ?
Numeric entry: NO
Depressing ENTR shall initiate the system test. After completion of the test, the current BIT status (see next item) shall be displayed.

Item 2 T O N O G O
Numeric entry: NO
Displays, from left to right, the TSCD, NSG, and GLLD error codes giving all test failures since power-up or the last "error clear" (see next item). Pressing ENTR shall have no effect.

Item 3 E R R O R C L R ?
Numeric entry: NO
Depressing ENTR shall clear the current BIT status error codes (preceding item) and accumulate any failures into the error recal item (see next item). The NSG self-test status display shall be reset. The TSCD, GLLD, and NSG fail indicators, along with the equipment and vehicle battery warning indicators, shall be turned off, unless there is a constant failure condition, in which case one or more lights may remain on. After executing the "error clear", the TSCD shall display "KEYBOARD RDY".

Item 4 T O N O G O
Numeric entry: NO
Displays, from left to right, the TSCD, NSG, and GLLD recall BIT status error codes giving the accumulated errors which have been cleared by the "error clear" item. Pressing ENTR shall have no effect.

TABLE IV. Error code definition.

Test number/BIT position	TSCD error code
0 [least significant (LS) bit]	ROM check
1	RAM check
2	GLLD and DMD interface
3	NSG interface
4	Data processing error
5	Display ROM
6	Display RAM
7	Watch Dog

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NSG error code

0 (LS bit)	UART error
1	Time-out
2	NSG self-test
3	NSG data type error
4	NSG mode error
5	NSG sends unintelligible data
6	Northings/Eastings compare test

GLLD error code

0 (LS bit)	GLLD fail bit
1	Time-out
2	GLLD code wrap
3	GLLD wrap - commanded
4	GLLD wrap - AZ-Adjust
5	GLLD data type error
6	GLLD parity bit

Item 5 N S G M F F S O
Numeric entry: NO
Displays the current NSG mode and self-test status. Pressing ENTR shall update the display. Permissible modes are as follows:

Mode number	Mode
FF	NSG power off mode
0	Start up mode
1	Coarse align mode
2	Fine align mode
3	Directional Gyro (DG) mode
4	Realign mode

NSG self-test status is provided in table III.

30.8 Control of G/VLLD functions. The TSCD shall control the operation of the G/VLLD, dependent upon the action and trigger switches on the targeting station hand control assembly. When either trigger is pulled, with the G/VLLD power on, a valid GLLD code entered, and no laser inhibit interrupts enabled, the TSCD shall command the G/VLLD to fire as follows:

a. When in a Range mode, the TSCD shall supply the G/VLLD with azimuth and elevation data in resolver form (per MIS-34260) before the G/VLLD generates a computer word. When a G/VLLD word is generated, it shall be intercepted by the TSCD, and the azimuth and elevation data shall be replaced with azimuth and elevation as received directly from the NSG. Range data from the G/VLLD shall be stored in the TSCD. The composite G/VLLD word shall be sent to the DMD with a recalculated parity.

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b. When in a Designate mode, no data is received from the G/VLLD. The TSCD shall keep the G/VLLD in the Designate mode for as long as the trigger is held.

TABLE V. Northseeking gyrocompass (NSG) self-test status.

HEXIDECIMAL TYPE NUMBER	FAILURE TYPE/NOMENCLATURE
00	Pass
01	Reserved
02	Reserved
03	Reserved
04	Reserved
05	Reserved
06	Reserved
07	Reserved
08	Excessive D-G to North Seeking Transfer Time
09	Excessive North Seeking to D-G Transfer Time
0A	Gyro Compass Fine Filter Saturation
0B	Reserved
0C	ROM Sum Check
0D	A/D-D/A Check
0E	Servo Loop Saturation
0F	Excessive Azimuth Gyro Temperature
10	Elevation Range
11	ITA Switch
12	RAM Read/Write
13	Cal Prom Sum Check
14	Temperature Saturation

NOTE: 00 Hex indicates Pass. Any other number indicates a failure has been encountered.

c. When the action switch on the hand control is activated, and the trigger is not also activated, the TSCD shall place the G/VLLD in the "AZ Adjust" mode. In this mode, the TSCD shall supply the G/VLLD with azimuth and elevation data in resolver form, at an update rate of four times per seconds. This data shall appear in the G/VLLD optics display.

d. When one or both trigger switches on the targeting station hand control assembly are depressed or fail during TSCD power-up, no laser interrupts exist, and the G/VLLD power is on, a flashing "TRIG SHORT" operator error message shall appear the the trigger indicator on the hand control assembly shall be illuminated. The G/VLLD shall not fire and the overrid function shall not allow firing of the G/VLLD. When the "TRIG SHORT" message is cleared, all TSCD functions shall be regained except for the G/VLLD fire.

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e. When one or both trigger switches fail in a shorted switch condition during a mission, and no laser interrupt conditions exist, the G/VLLD shall fire once, if in the range mode, and no operator-TSCD keyboard interface shall be allowed. The G/VLLD shall fire continuously, if in the designate mode, until the G/VLLD power switch is turned off. The TSCD shall regain all functions, except the G/VLLD fire, when the G/VLLD power is turned off. Turning the G/VLLD power switch back on shall fire the G/VLLD in either the range or designate mode, allowing a degraded mode of operation when a trigger switch has failed.

30.9 Built-in-test (BIT). There shall be three categories of BIT incorporated in the TSCD software.

30.9.1 TSCD BIT. The TSCD tests shall include ROM (read only memory) and RAM (random access memory) tests, G/VLLD, NSG, and DMD interface tests, data processing error tests (e.g., errors occurring in number conversion routines due to errors in data types), and "watch dog" which is a circuit which shall reset the microporcessor if it does not receive a reset pulse from the microprocessor every 16.7 seconds.

30.9.2 NSG BIT. The NSG tests shall include the following:

a. UART - Checks for parity, framing, and overrun errors.

b. Time-out - If an NSG response was expected but not received within 50 ms (RTS active to CLS active or loss of CLS -25 msec) interval, a time-out error shall be flagged.

c. NSG self-test - The TSCD shall monitor the results of the NSG's own self-tests.

d. NSG data type error - This error occurs if the NSG sends "bad" numerical data (i.e., illegal characters).

e. NSG mode error - The TSCD shall flag as an error any improper mode sequencing of the NSG.

f. Undefined responses - Any response to a command which is undefined shall result in an error.

g. Northings/Eastings comparison - Northings/Eastings data is requested from the NSG and compared to the TSCD's data. An error shall be flagged if the data does not match.

30.9.3 G/VLLD tests. The G/VLLD tests shall include the following:

a. Fail Bit - The G/VLLD's fail BIT for low energy shall be checked during Ranging (during designating, the low energy fault must be detected through the G/VLLD eyepiece display indicator lamp illumination).

b. Time-out - This error shall occur if an expected G/VLLD response is not received within 200 ms.

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c. G/VLLD code - A wrap check shall be made of the G/VLLD code lines once every 10 seconds.

d. G/VLLD parity BIT - The parity of G/VLLD words shall be checked by the TSCD four times a second when G/VLLD is in AZ-Adjust or Range mode.

e. G/VLLD data type - Numeric data from the G/VLLD shall be checked for illegal data types.

f. G/VLLD word wraparound - When in the Azimuth Adjust mode, azimuth and elevation data received from the G/VLLD shall be compared to what was sent.

30.9.4 Frequency of tests. All TSCD BIT tests shall be performed at power-up. In addition, ROM and RAM tests shall be performed on a continuous basis when the operator is not interfacing with the TSCD. G/VLLD and NSG tests shall also be performed applicable to their modes of operation. While the NSG is on, mode and status shall be requested by the TSCD at least every 10 seconds. An operator-initiated "system-test" shall perform all of the TSCD tests, as well as G/VLLD and NSG interface tests.

30.10 Operator error messages. The TSCD shall display an operator error message when an attempt is made to enter an out-of-range number, or when certain operations are requested which cannot be carried out. All error messages except "NSG NOT RDY" shall clear only upon pressing the CLR key. Table III contains a list of the error messages and their meanings.

30.11 Indicators.

30.11.1 Targeting head status indicators. The targeting head status indicators are used to prompt the targeting station operator in the operation of the targeting head. The indicators shall be as specified in table VI.

30.11.2 Caution and fault indicators. The illumination of an indicator in this group shall alert the operator that the BIT has detected a fault in the TSCD, NSG, or G/VLLD; or that the vehicle or equipment battery voltage is low. These indicators shall remain illuminated until cleared by the operator using the ERROR CLR item of the TEST key menu (see 30.7.10). For BIT faults, the TEST key shall be used to read out an error code describing the fault in more detail.

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TABLE VI. TSCD indicator lights.

Indicator	Function
CARGO HATCH	This indicator shall illluminate when the FISTV cargo hatch is not securely closed, causing a laser fire/targeting station motion interrupt condition to exist.
WPN HATCH	This indicator shall illuminate when the weapon hatch is not securely closed, causing a laser fire/targeting station motion interrupt condition to exist.
DRIVER HATCH	This indicator shall illuminate when the FISTV driver's hatch is not securely closed, causing a laser fire/targeting station motion interrupt condition to exist.
LASER INTRPT	This indicator shall illuminate when a laser interrupt condition exists, which will prevent the G/VLLD from firing.
EL ABOVE	This indicator shall illuminate when the targeting head is above the proper elevation for stowing.
EL STOW	This indicator shall illuminate when the targeting head is at the proper elevation for stowing.
EL BELOW	This indicator shall illuminate when the targeting head is below the proper elevation for stowing.
HEAD STOW	This indicator shall illuminate when the targeting head is in stow position.
AZ STOW	This indicator shall illuminate when the targeting head is in the proper azimuth position for stowing.
HEAD ERECT	This indicator shall illuminate when the targeting head is in the full erect position and erection arm locks are engaged.
EQP BAT	This indicator shall illuminate when the voltage across the equipment battery falls below the 21 $\pm$ 0.5 V for a period greater than 500 ms.
VEH BAT	This indicator shall illuminate when the voltage across the vehicle battery falls below the 17.5 $\pm$ 0.5 V for a period greater than 500 ms.
HEAD READY	This indicator shall illuminate when the targeting head is fully erect with both the erection arm and G/VLLD locks engaged.

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TABLE VI. TSCD indicator lights - Continued.

Indicator	Function
NSG ON	This indicator shall illuminate when the TSCD PWR and NSG switches are both in the ON position. This puts a +24 V signal on the NSG PWR CMD output line.
NSG FAULT	This indicator shall illuminate when a fault is detected by the BIT of the NSG or TSCD.
NSG ALIGN	This indicator shall be illuminated prior to initialization of NSG at system turn-on and during initialization, reinitialization, and realignment. If no valid vehicle coordinates have been entered, the lamp will flash at a rate of twice per second until valid coordinates are entered.
NSG READY	This indicator shall illuminate when the NSG has been aligned and is in the Directional Gyro (DG) mode.
TSCD FAULT	This indicator shall illuminate when a fault is detected in the TSCD circuitry during testing.
GLLD FAULT	This indicator shall illuminate when a fault is detected during testing of the GLLD interfacing circuitry.
GLLD ON	This indicator shall illuminate when the GLLD switch is in the ON position. This puts a +24 V signal on the GLLD PWR CMD output line.

30.11.3 Interrupt status indicators. The laser interrupt, cargo hatch, driver's hatch, and weapon hatch interrupt indicators shall illuminate to signal an interrupt condition which will inhibit laser fire and/or turret motion. Interrupts may be overridden by use of the interrupt override switch on the TSCD control panel as specified in table VII. The wide-field-of-view (WFOV) selected and stow position conditions shall not be overridden.

3.12 Control switches. The control switches, listed in table VIII, shall be located on the TSCD front panel as shown in figure 1. The NSG power and TSCD power switches shall be positive action guarded, and the pump disable and interrupt override switches covered to prevent accidental activation/deactivation.

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TABLE VII. Targeting station safety interrupt conditions.

CONDITION	FUNCTION ENABLED		
	LASER FIRE	STOW	TURRET MOTION*
Driver's Hatch open	No #	Yes	No #
Targeting Station Hatch Open	No #	Yes	No #
Cargo Hatch Open (Popped up)	No #	No	No #
Cargo Hatch Full Open	No #	Yes	Yes
Stow Position	No	N/A	No
G/VLLD Mount Unlocked	Yes	Yes	No #
Erect Locks Not Engaged	Yes	Yes	No #
No Azimuth Stow	Yes	No	Yes
WFOV Selected	No	Yes	Yes
Trigger Switch Shorted (during power-up)	No	Yes	Yes
Trigger Switch Shorted (during mission)	Yes	Yes	Yes

* - Turret motion consists of track, slew, and elevation.
- Indicates override will enable function.

TABLE VIII. TSCD control switches.

Switch	Function
TSCD PWR	This switch supplies power to the TSCD and the turret. When this switch is in the ON position, the equipment battery voltage is applied to the TSCD PWR SW and TUR PWR CMD lines. This switch shall be positive action guarded to prevent accidental activation/deactivation.
NSG	This switch supplies power to the NSG. When both the TSCD PWR and NSG switches are placed in the ON position, +24 V dc is applied to the NSG PWR CMD line.
GLLD	This switch supplies power to the G/VLLD and selects the operating mode of G/VLLD. While the switch is in the DES position, the G/VLLD operates as a continuous-pulsed laser designator. The RNG1 position is used for determining position of hard targets, and the RNG2 position is used for determining position of soft targets.
HEAD	This switch selects the mode of operation of the targeting head (ERECT or STOW) erection drive motor.

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TABLE VIII. TSCD control switches -Continued.

Switch	Function
DRIVE	This switch activates the erection drive motor, when held to the ON position, causing the targeting head to move to the position selected by HEAD switch. The DRIVE switch is spring-loaded to the OFF position.
INTERRUPT	This switch causes specified interrupt conditions to be overridden. The swtich must be in the OVERRIDE position to override in the interrupts noted in table III. This switch shall be covered to prevent accidental activation.
PUMP DISABLE	This switch removes power from the targeting station hydraulic pump. The switch shall be covered to prevent accidental activation. It shall normally be used only when servicing the hydraulic system, or as an emergency shutoff in the event of a hydraulic system failure.

40. QUALITY ASSURANCE PROGRAMS

40.1 Method of inspection. The examinations and tests for key/switch functions and indicators of TBCD units depicted in 40.1 (inclusive) are the minimum required to determine conformance to the requirements delineated in 30.1 of this appendix. Additional examinations and tests by the contractor may be required to determine conformance to specification requirements. The Government reserves the authority to conduct the inspections/tests depicted in 40.1 (inclusive) and additional inspections/tests at the discretion of the Government to determine conformance of end items or components to specification requirements. The TSCD shall be tested as specified as followings in section 40.1 (inclusive).

40.1.1 Initial input conditions. The following initial input conditions shall be present before performing any test or examination specified in section 40.1 (inclusive). Prime power (+24 V dc) shall be applied to pins J1-H, J1-w, and J1-z (see figure 4).

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J1 Connector

A	SPARE	AA	GND
B	SPARE	BB	WFOVC
C	SPARE	CC	WFOVS
D	SPARE	DD	ACT LT
E	SPARE	EE	ACT RT
F	SPARE	FF	SLEW
G	SPARE	GG	TRIG RT
H	TUR PWR	HH	TRIG LT
J	NSG PWR SW		
K	GLLD PWR SW		
L	TRIG IND RTN		
M	GND		
N	HND CRNK		
P	HYD PRESS SW		
R	SLW BRK SOL		
S	TRK BRK SOL		
T	EL DR SOL		
U	WPN HTCH		
V	EL STOW		
W	BLW STOW		
X	ABV STOW		
Y	STOW CONF		
Z	ERCT CONF		
a	CW DR SOL		
b	ERCT LKS		
c	GLLD +24		
d	GLLD CVR		
e	GLLD LKS		
f	GLLD LK SOL		
g	BYPS SOL		
h	ERCT LK SOL		
i	CCW DR SOL		
j	(DMD WORD)		
k	DESIGNATE COND		
m	DMD RTN		
n	DRV HTCH		
p	CARGO HTCH OPN		
q	CARGO HTCH FOPN		
r	AZ STOW		
s	AUD ALRT		
t	BAT GND		
u	TURRET PWR CMD		
v	HYD RLY RTN		
w	TSCD PWR		
x	GLLD PWR CMD		
y	NSG PWR CMD		
z	VEH BAT +24		

J2 Connector

A	SPARE	AA	(NSG RTS)
B	SPARE	BB	(NSG RTS)
C	SPARE	CC	(NSG CTS)
D	SPARE	DD	(NSG CTS)
E	GLLD CODE 1	EE	(NSG BIT)
F	GLLD CODE 2	FF	(NSG BIT)
G	GLLD CODE 3	GG	SPARE
H	GLLD CODE 4	HH	SPARE
J	GLLD CODE 5		
K	GLLD CODE 6		
L	GLLD CODE 7		
M	GLLD CODE 8		
N	GLLD CODE 9		
P	GLLD RTN		
R	RNG1/RNG2		
S	AZ ADJ		
T	FISTV DES		
U	RET BRT		
V	TSCD PWR SW		
W	KFIRE		
X	SPARE		
Y	SPARE		
Z	SPARE		
a	SPARE		
b	FIRE CMD		
c	(GLLD WORD)		
d	(GLLD WORD)		
e	FIRE REMOTE		
f	GND		
g	GND		
h	(NSG SDI)		
i	(NSG SDI)		
j	GND		
k	EL SIN		
m	EL COS		
n	EL COM		
p	AZ SIN		
q	AZ COS		
r	AZ COM		
s	RSLVR REF		
t	RSLVR RTN		
u	SPARE		
v	SPARE		
w	(NSG SDO)		
x	(NSG SDO)		
y	SPARE		
z	SPARE		

FIGURE 4. TSCD J1, J2 connector pin designations.

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INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
TSCD PWR (ON/OFF)	PANEL	OFF		
GLLD	PANEL	OFF		
PUMP (ON/DISABLE)	PANEL	ON		
INTERRUPT	PANEL	Not Override		
HEAD (Erect/Stow)	PANEL	Stow		
DRIVE (ON/OFF)	PANEL	OFF		
NSG (ON/OFF)	PANEL	OFF		
(Control Panel Switches)				
VEH GRID	SW1	OFF		
TGT GRID	SW2	OFF		
1	SW3	OFF		
2	SW4	OFF		
3	SW5	OFF		
HDG	SW6	OFF		
4	SW8	OFF		
5	SW9	OFF		
6	SW10	OFF		
DMD WORD	SW11	OFF		
GLLD CODE	SW12	OFF		
7	SW13	OFF		
8	SW14	OFF		
9	SW15	OFF		
CALC	SW16	OFF		
CLR	SW17	OFF		
+/-	SW18	OFF		
0	SW19	OFF		
ENTR	SW20	OFF		
LAMP TST BRT	SW21	OFF		
GLLD RET BRT	SW23	OFF		
NSG ALIGN	SW24	OFF		
TEST	SW25	OFF		
NSG PW SW	J1-J	OPEN		
GLLD PW SW	J1-K	OPEN		
HND CRNK	J1-N	OPEN		
HYD PRESS SW	J1-P	OPEN		
WPN HTCH	J1-U	CLOSED		
EL STOW	J1-V	OPEN		
BLW STOW	J1-W	OPEN		
ABV STOW	J1-X	OPEN		
STOW CONF	J1-Y	OPEN		
ERECT CONF	J1-Z	CLOSED		
ERCT LKS	J1-b	OPEN		
GLLD CVR	J1-d	OPEN		
GLLD LKS	J1-e	OPEN		
DMD RTN	J1-m	CLOSED		
DRVR HTCH	J1-n	CLOSED		

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	INPUT SIGNALS		OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
CRGO HTCH OPN	J1-p	CLOSED		
CRGO HTCH FOPN	J1-q	OPEN		
AZ STOW	J1-r	OPEN		
BAT GND	J1-t	CLOSED		
WFOVC	J1-BB	OPEN		
WFOVS	J1-CC	CLOSED		
ACT LT	J1-DD	OPEN		
ACT RT	J1-EE	OPEN		
SLEW	J1-FF	OPEN		
TRIG RT	J1-GG	OPEN		
TRIG LT	J1-HH	OPEN		
GLLD RTN	J2-P	CLOSED		
GLLD WORD	J2-c	IDLE		
GLLD WORD	J2-d	IDLE		
RSLVR REF	J2-s	CLOSED		
RSLVR RTN	J2-t	CLOSED		
NSG SDI	J2-y	IDLE		
NSG SDI	J2-z	IDLE		
NSG CTS	J2-CC	IDLE		
NSG CTS	J2-DD	IDLE		
NSG BIT	J2-EE	IDLE		
NSG BIT	J2-FF	IDLE		
TSCD PWR	J1-w	CLOSED (24 ± 1 V dc)		
TUR PWR	J1-H	CLOSED (24 ± 1 V dc)		
VEH BAT	J1-z	CLOSED (24 ± 1 V dc)		
TSCD PWR (ON/OFF)	PANEL	ON	DISPLAY	flashing " STANDBY "

40.1.2 Display. To determine conformance to 30.3, it shall be verified that up to twelve alphanumeric characters are displayed on the TSCD and that the characters displayed agree with the input characters or the proper error message. See table III for the allowable error messages.

40.1.3 ENTR (enter) key test. To verify conformance to 30.5, it shall be demonstrated that pressing ENTR terminates the entry of a numeric string and displays the next menu item if the entry is a valid number, or displays "KEYBOARD RDY" if it is the last item. A non-numeric entry (i.e, a function key) shall be depressed followed by the ENTR key and it shall be verified that the specific function is performed. It shall also be shown that the ENTR key has no effect when an operator error message is displayed.

40.1.4 CLR (clear) key test. To verify conformance to 30.6, it shall be shown that the CLR key clears an operator error message and displays the previous menu item, discards a new numeric entry and displays the previous number, or if neither of the above, causes "KEYBOARD RDY" to be displayed.

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40.1.5 VEH GRID (vehicle grid) key test. To verify conformance to 30.1.1, the following sequence of operations shall be performed and the appropriate output signals demonstrated.

INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
VEH GRID	SW1	ON	DISPLAY	V EN########
VEH GRID	SW1	ON	DISPLAY	V E ######
VEH GRID	SW1	ON	DISPLAY	V N ########
VEH GRID	SW1	ON	DISPLAY	V A ####
VEH GRID	SW1	ON	DISPLAY	KEYBOARD RDY
VEH GRID	SW1	ON	DISPLAY	V EN########
VEH GRID	SW1	ON	DISPLAY	V E ######
1	SW3	ON	DISPLAY	"V E 1"
2	SW4	ON	DISPLAY	"V E 12"
3	SW5	ON	DISPLAY	"V E 123"
4	SW8	ON	DISPLAY	"V E 1234"
5	SW9	ON	DISPLAY	"V E 12345"
6	SW10	ON	DISPLAY	"V E 123456"
7	SW13	ON	DISPLAY	" TOO MANY "
CLR	SW17	ON	DISPLAY	"V E ######"
7	SW13	ON	DISPLAY	"V E 7"
8	SW14	ON	DISPLAY	"V E 78"
9	SW15	ON	DISPLAY	"V E 789"
+/-	SW18	ON	DISPLAY	"V E -789"
0	SW19	ON	DISPLAY	"V E -7890"
CLR	SW17	ON	DISPLAY	"V E ######"
1	SW3	ON	DISPLAY	"V E 1"
1	SW3	ON	DISPLAY	"V E 11"
9	SW15	ON	DISPLAY	"V E 119"
9	SW15	ON	DISPLAY	"V E 1199"
9	SW15	ON	DISPLAY	"V E 11199"
9	SW15	ON	DISPLAY	"V E 119999"
ENTR	SW20	ON	DISPLAY	"? E <122 KM"
CLR	SW17	ON	DISPLAY	"V E "
ENTR	SW20	ON	DISPLAY	"? V EASTINGS"
CLR	SW17	ON	DISPLAY	"V E ######"
8	SW14	ON	DISPLAY	"V E 8"
7	SW13	ON	DISPLAY	"V E 87"
8	SW14	ON	DISPLAY	"V E 878"
0	SW19	ON	DISPLAY	"V E 8780"
0	SW19	ON	DISPLAY	"V E 87800"
1	SW3	ON	DISPLAY	"V E 878001"
ENTR	SW20	ON	DISPLAY	"? E >878 KM"
CLR	SW17	ON	DISPLAY	"V E ######"

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INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
VEH GRID	SW1	ON	DISPLAY	"V N ########'
1	SW3	ON	DISPLAY	"V N 1'
0	SW19	ON	DISPLAY	"V N 10'
0	SW19	ON	DISPLAY	"V N 100'
0	SW19	ON	DISPLAY	"V N 1000'
0	SW19	ON	DISPLAY	"V N 10000'
0	SW19	ON	DISPLAY	"V N 100000'
0	SW19	ON	DISPLAY	"V N 1000000'
0	SW19	ON	DISPLAY	"V N 10000000'
ENTR	SW20	ON	DISPLAY	"N OUT OF RNG'
CLR	SW17	ON	DISPLAY	"V N ########'
ENTR	SW20	ON	DISPLAY	"? V NORTHING'
CRL	SW17	ON	DISPLAY	"V N ########'
VEH GRID	SW1	ON	DISPLAY	"V A ####'
+/-	SW18	ON	DISPLAY	"V A -'
4	SW8	ON	DISPLAY	"V A -4'
0	SW19	ON	DISPLAY	"V A -40'
1	SW3	ON	DISPLAY	"V A -401'
ENTR	SW20	ON	DISPLAY	"? ALT <-400M'
CRL	SW17	ON	DISPLAY	"V A ####'
7	SW13	ON	DISPLAY	"V A 7'
5	SW9	ON	DISPLAY	"V A 75'
0	SW19	ON	DISPLAY	"V A 750'
1	SW3	ON	DISPLAY	"V A 7501'
ENTR	SW20	ON	DISPLAY	"? ALT > 7500'
CLR	SW17	ON	DISPLAY	"V A ####"
ENTR	SW20	ON	DISPLAY	"? V ALTITUDE"
CLR	SW17	ON	DISPLAY	"V A ####"
VEH GRID	SW1	ON	DISPLAY	"KEYBOARD RDY"

40.1.6 TGT GRID (target grid) key test. To verify conformance to 30.7.2, the following sequence of operations shall be performed and the appropriate output signals demonstrated.

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INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
TGT GRID	SW2	ON	DISPLAY	"T EN########"
TGT GRID	SW2	ON	DISPLAY	"T E ######"
TGT GRID	SW2	ON	DISPLAY	"T N ########"
TGT GRID	SW2	ON	DISPLAY	"T A ####"
TGT GRID	SW2	ON	DISPLAY	"KEYBOARD RDY"
TGT GRID	SW2	ON	DISPLAY	"T EN########"
TGT GRID	SW2	ON	DISPLAY	"T E ######"
1	SW3	ON	DISPLAY	"T E 1"
1	SW3	ON	DISPLAY	"T E 11"
9	SW15	ON	DISPLAY	"T E 119"
9	SW15	ON	DISPLAY	"T E 1199"
9	SW15	ON	DISPLAY	"T E 11199"
9	SW15	ON	DISPLAY	"T E 119999"
ENTR	SW20	ON	DISPLAY	"? E <122 KM"
CLR	SW17	ON	DISPLAY	"T E ######"
ENTR	SW20	ON	DISPLAY	"? T EASTINGS"
CLR	SW17	ON	DISPLAY	"T E ######"
TGT GRID	SW2	ON	DISPLAY	"T N ########"
1	SW3	ON	DISPLAY	"T N 1"
0	SW19	ON	DISPLAY	"T N 10"
0	SW19	ON	DISPLAY	"T N 100"
0	SW19	ON	DISPLAY	"T N 1000"
0	SW19	ON	DISPLAY	"T N 10000"
0	SW19	ON	DISPLAY	"T N 100000"
0	SW19	ON	DISPLAY	"T N 1000000"
0	SW19	ON	DISPLAY	"T N 10000000"
ENTR	SW20	ON	DISPLAY	"N OUT OF RNG"
CLR	SW17	ON	DISPLAY	"T N ########"
ENTR	SW20	ON	DISPLAY	"? T NORTHING"
CRL	SW17	ON	DISPLAY	"T N ########"
TGT GRID	SW2	ON	DISPLAY	"T A ####"
7	SW13	ON	DISPLAY	"T A 7"
5	SW9	ON	DISPLAY	"T A 75"
0	SW19	ON	DISPLAY	"T A 750"
1	SW3	ON	DISPLAY	"T A 7501"
ENTR	SW20	ON	DISPLAY	"? ALT > 750M"
CLR	SW17	ON	DISPLAY	"T A ####"
ENTR	SW20	ON	DISPLAY	"? V ALTITUDE"
CLR	SW17	ON	DISPLAY	"T A ####"
TGT GRID	SW2	ON	DISPLAY	"KEYBOARD RDY"

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40.1.7 HDG (heading) key test. To verify conformance to 30.7.5, the following sequence of operations shall be performed and the appropriate output signals demonstrated. With the turret-vehicle simulator simulating a stowed targeting head, depress the HDG key and verify a display "H V ####". With the turret-vehicle simulator simulating an erected targeting head, depress the HDG key and verify a message of "NSG NOT READY" if the NSG has not been aligned or a message of "H T #### ###".

40.1.8 DMD WORD key test. To verify conformance to 30.7.3, the following sequence of operations shall be performed and the appropriate output signals demonstrated.

INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
DMD WORD	SW11	ON	DISPLAY	" 0 999 0"
DMD WORD	SW11	ON	DISPLAY	"DIR 0"
6	SW11	ON	DISPLAY	"DIR 6"
4	SW8	ON	DISPLAY	"DIR 64"
0	SW19	ON	DISPLAY	"DIR 640"
0	SW20	ON	DISPLAY	"DIR 6400"
ENTR	SW20	ON	DISPLAY	"AZ > 6399MIL"
CLR	SW17	ON	DISPLAY	"DIR 0"
DMD WORD	SW11	ON	DISPLAY	"SLT DIST 999"
3	SW5	ON	DISPLAY	"SLT DIST 3"
9	SW15	ON	DISPLAY	"SLT DIST 39"
ENTR	SW20	ON	DISPLAY	"RNG < 400 M "
CLR	SW27	ON	DISPLAY	"SLT DIST 999"
DMD WORD	SW11	ON	DISPLAY	"VA 0"
1	SW3	ON	DISPLAY	"VA 1"
0	SW19	ON	DISPLAY	"VA 10"
0	SW19	ON	DISPLAY	"VA 100"
0	SW19	ON	DISPLAY	"VA 1000"
ENTR	SW20	ON	DISPLAY	"OUT OF RANGE"
CLR	SW17	ON	DISPLAY	"VA 0"
DMD WORD	SW11	ON	DISPLAY	"SEND (ENTER)"
DMD WORD	SW11	ON	DISPLAY	"KEYBOARD RDY"

40.1.9 GLLD CODE key test. To verify conformance to 30.7.6, the following sequence of operations shall be performed and the appropriate output signals demonstrated.

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INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
GLLD	SW12	ON	DISPLAY	"GLLDCODE ###"
1	SW3	ON	DISPLAY	"GLLDCODE 1"
1	SW3	ON	DISPLAY	"GLLDCODE 11"
0	SW3	ON	DISPLAY	"GLLDCODE 110"
ENTR	SW20	ON	DISPLAY	" ? GLLD CODE"
CLR	SW17	ON	DISPLAY	"GLLDCODE ###"
8	SW14	ON	DISPLAY	"GLLDCODE 8"
8	SW14	ON	DISPLAY	"GLLDCODE 88"
9	SW15	ON	DISPLAY	"GLLDCODE 889"
ENTR	SW20	ON	DISPLAY	" ? GLLD CODE"
CLR	SW17	ON	DISPLAY	"GLLDCODE ###"
GLLD CODE	SW12	ON	DISPLAY	"KEYBOARD RDY"

40.1.10 CALC (calculate key test). To verify conformance to 30.7.4, the following sequence of operations shall be performed and the appropriate output signals demonstrated.

INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
CALC	SW16	ON	DISPLAY	"C V LAST TGT"
CALC	SW16	ON	DISPLAY	"C T LAST TGT"
CALC	SW16	ON	DISPLAY	"KEYBOARD RDY"
CALC	SW16	ON	DISPLAY	"C V LAST TGT"
CLR	SW17	ON	DISPLAY	"KEYBOARD RDY"

40.1.11 LAMP TEST/BRT (lamp test/brightness) key test. To verify conformance to 30.7.7, the following sequence of operations shall be performed and the appropriate output signals demonstrated. It shall be verified that the lamp brightness changes proportionally to the inputed value with 9 being the maximum intensity and 1 the minimum intensity.

INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
LAMP TEST/BRT	SW21	ON	DISPLAY	"LAMP BRT 9"
1	SW3	ON	DISPLAY	"LAMP BRT 1"
ENTR	SW20	ON	DISPLAY	"KEYBOARD RDY" (All illuminated lamps and displays dim to minimum intensity).

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INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
LAMP TEST/BRT	SW21	ON	DISPLAY	"LAMP BRT 1"
2	SW4	ON	DISPLAY	"LAMP BRT 2"
ENTR	SW20	ON	DISPLAY	"KEYBOARD RDY" (All illuminat lamps and dis- plays brighten
LAMP TEST/BRT	SW21	ON	DISPLAY	"LAMP BRT 2"
4	SW8	ON	DISPLAY	"LAMP BRT 4"
ENTR	SW20	ON	DISPLAY	"KEYBOARD RDY" (All illuminat lamps and dis- plays brighten
LAMP TEST/BRT	SW21	ON	DISPLAY	"LAMP BRT 4"
6	SW10	ON	DISPLAY	"LAMP BRT 6"
ENTR	SW20	ON	DISPLAY	"KEYBOARD RDY" (All illuminat lamps and dis- plays brighten
LAMP TEST/BRT	SW21	ON	DISPLAY	"LAMP BRT 6"
9	SW15	ON	DISPLAY	"LAMP BRT 9"
ENTR	SW20	ON	DISPLAY	"KEYBOARD RDY" (All illuminat lamps and dis- plays brighten to maximum intensity.)

40.1.12 GLLD RET BRT (GLLD reticle brightness) key test. To verify conformance to 30.7.8, the following sequence of operations shall be performed and the appropriate output signals demonstrated. It shall be verified that the GLLD reticle brightness voltages change proportional to the input value with 9 being the maximum intensity and 1 the minimum intensity.

INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
GLLD RET BRT	SW23	ON	DISPLAY J2-U	"GLLD RET 1 0.5 V dc max. (RTCL BRT)
2	SW4	ON	DISPLAY	"GLLD RET 2
ENTR	SW20	ON	DISPLAY J2-U	"KEYBOARD RDY 2.6 - 3.4 V d (RTCL BRT)

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INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
GLLD RET BRT	SW23	ON	DISPLAY	"GLLD RET 2"
3	SW5	ON	DISPLAY	"GLLD RET 3"
ENTR	SW20	ON	DISPLAY	"KEYBOARD RDY"
			J2-U	3.9 - 4.7 V dc (RTCL BRT)
GLLD RET BRT	SW23	ON	DISPLAY	"GLLD RET 3"
4	SW8	ON	DISPLAY	"GLLD RET 4"
ENTR	SW20	ON	DISPLAY	"KEYBOARD RDY"
			J2-U	4.8 - 5.6 V dc (RTCL BRT)
GLLD RET BRT	SW23	ON	DISPLAY	"GLLD RET 4"
5	SW9	ON	DISPLAY	"GLLD RET 5"
ENTR	SW20	ON	DISPLAY	"KEYBOARD RDY"
			J2-U	5.3 - 5.9 V dc (RTCL BRT)
GLLD RET BRT	SW23	ON	DISPLAY	"GLLD RET 5"
6	SW10	ON	DISPLAY	"GLLD RET 6"
ENTR	SW20	ON	DISPLAY	"KEYBOARD RDY"
			J2-U	6.0 - 6.6 V dc (RTCL BRT)
GLLD RET BRT	SW23	ON	DISPLAY	"GLLD RET 6"
7	SW13	ON	DISPLAY	"GLLD RET 7"
ENTR	SW20	ON	DISPLAY	"KEYBOARD RDY"
			J2-U	6.5 - 7.1 V dc (RTCL BRT)
GLLD RET BRT	SW23	ON	DISPLAY	"GLLD RET 7"
8	SW14	ON	DISPLAY	"GLLD RET 8"
ENTR	SW20	ON	DISPLAY	"KEYBOARD RDY"
			J2-U	6.9 - 7.5 V dc (RTCL BRT)
GLLD RET BRT	SW23	ON	DISPLAY	"GLLD RET 8"
9	SW15	ON	DISPLAY	"GLLD RET 9"
ENTR	SW20	ON	DISPLAY	"KEYBOARD RDY"
			J2-U	7.0 - 7.6 V dc (RTCL BRT)

40.1.13 NSG ALIGN key test. To verify conformance to 30.7.9, the following sequence of operations shall be performed and the appropriate output signals demonstrated.

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INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
NSG ALIGN	SW24	ON	DISPLAY	"NSG RE-ALIGN"
ENTR	SW20	ON	DISPLAY	"NO VALID E/N"
CLR	SW17	ON	DISPLAY	"NSG RE-ALIGN"
NSG ALIGN	SW24	ON	DISPLAY	"NSG RE-INIT"
ENTR	SW20	ON	DISPLAY	"NO VALID E/N"
CLR	SW17	ON	DISPLAY	"NSG RE-INIT"
NSG ALIGN	SW24	ON	DISPLAY	"KEYBOARD RDY"

40.1.14 TEST key test. To verify conformance to 30.7.10, the following sequence of operations shall be performed and the appropriate output signals demonstrated.

INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
TEST	SW25	ON	DISPLAY	"SYSTEM-TEST"
TEST	SW25	ON	DISPLAY	"T O N O G O"
TEST	SW25	ON	DISPLAY	"ERROR CLR? "
TEST	SW25	ON	DISPLAY	"T O N40 G 2"
TEST	SW25	ON	DISPLAY	"NSG MFF S O"
TEST	SW25	ON	DISPLAY	"KEYBOARD RDY"

40.1.15 Trigger short test. To verify conformance to 30.8, the following sequence of operations shall be performed and the appropriate output signals demonstrated.

INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
HEAD (ERECT/STOW)	PANEL	ERECT	J1-h	1.5 V dc (max) (ERCT LK SOL)
ERCT LKS	J1-b	CLOSED	LAMPS	HEAD ERECT - illuminated; NSG ON - illuminated; NSG RDY - illuminated; No other lamps illuminated
			J1-E	1.5 V dc (max) (GLLD LK SOL)

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INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
GLLD LKS	J1-e	CLOSED	LAMPS	HEAD ERECT - illuminated; HEAD RDY - illuminated; NSG ON - illuminated; NSG RDY - illuminated
TSCD PWR (ON/OFF)	PANEL	OFF		
GLLD	PANEL	RNG2		
TRIG RT	J1-GG	CLOSED	J2-V	24.0 ± 1 V dc (KFIRE +24)
TSCD PWR (ON/OFF)	PANEL	ON		
TRIG RT	J1-GG	OPEN	DISPLAY	"STANDBY" (flashes 6 times)
			DISPLAY	"TRIG SHORT" (flashing)
TRIG RT	J1-GG	CLOSED	J2-v	24.0 ± 1 V dc (KFIRE +24)
			J2-e	0.5 + 0.5 V dc (KFIRE REMOTE)
			J1-L	0.5 ± 0.5 V dc (TRIG IND)
TRIG RT	J1-GG	OPEN		
CLR	SW17	ON	DISPLAY	"V E ######"

40.1.16 BIT verification. The BIT is software based testing which is functionally verified during the TEST key tests (see 40.1.14). Conformance to 3.2.1.7 shall be shown by successfully accumulating and displaying errors during the TEST key operations.

40.1.17 Indicators and controls examination. To verify conformance to 30.11 and 30.12, the TSCD controls and indicators shall be examined proper operation and conformance to table VI.

40.1.18 Interrupt tests. To determine conformance to 3.11.1, table VII shall be verified along with the performance of the following tests.

a. Stow interrupt check. The following procedure shall be performed to verify that for the targeting head in the stowed mode, the laser cannot be fired and turret motion is inhibited. It shall also verify that the override function does not enable firing of the laser, nor movement of the targeting head, and sounds the alarm.

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INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
HEAD	PANEL	STOW		
ERECT LKS	J1-b	OPEN		
GLLD LKS	J1-e	OPEN		
STOW CONF	J1-y	24.0 ± 1 V dc	LAMPS	HEAD STOW - illuminated
GLLD	PANEL	RNG2	J2-R	5.0 ± 0.1 V dc
				$\overline{\text{(RNG1/RNG2)}}$
TRIG RT	J1-GG	24.0 ± 1 V dc	J2-W	0.5 ± 0.1 V dc
				$\overline{\text{(KFIRE)}}$
			J2-S	0.5 ± 0.1 V dc
				$\overline{\text{(AZ ADJ)}}$
			J2-e	5.0 ± 0.1 V dc
				$\overline{\text{(FIRE REMOTE)}}$
			J1-L	0.5 ± 0.1 V dc
				$\overline{\text{(TRIG IND)}}$
			LAMPS	LASER INTR - illuminated
TRIG RT	J1-GG	OPEN	J2-e	0.5 ± 0.1 V dc
				$\overline{\text{(FIRE REMOTE)}}$
			J1-L	24.0 ± 1 V dc
				(TRIG IND)
			LAMPS	LASER INTR - not illuminated
SLEW	J1-FF	24.0 ± 1 V dc	J2-S	5.0 ± 0.1 V dc
				(AZ ADJ)
			J1-S	24.0 ± 1 V dc
				$\overline{\text{(TRK BRK SOL)}}$
			J1-R	24.0 ± 1 V dc
				$\overline{\text{(SLW BRK SOL)}}$
			J1-T	24.0 ± 1 V dc
				$\overline{\text{(EL DR SOL)}}$

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INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
SLEW	J1-FF	OPEN	J1-S	24.0 ± 1 V dc
				(TRK BRK SOL)
			J1-R	24.0 ± 1 V dc
				(SLW BRK SOL)
			J1-T	24.0 ± 1 V dc
				(EL DR SOL)
INTERRUPT	PANEL	OVERRIDE	J1-s	24.0 ± 1 V dc
				(AUD ALERT)
TRIG RT	J1-GG	24.0 ± 1 V dc	J2-W	0.5 ± 0.1 V dc
				(KFIRE)
			J2-S	0.5 ± 0.1 V dc
				(AZ ADJ)
			J2-e	5.0 ± 0.1 V dc
				(FIRE REMOTE)
			J1-L	0.5 ± 0.1 V dc
				(TRIG IND)
			LAMPS	LASER INTR - illuminated
TRIG RT	J1-GG	OPEN	J2-e	0.5 ± 0.1 V dc
				(FIRE REMOTE)
			J1-L	24.0 ± 1 V dc
				(TRIG IND)
SLEW	J1-FF	24.0 ± 1 V dc	J1-S	24.0 ± 1 V dc
				(TRK BRK SOL)
			J1-R	24.0 ± 1 V dc
				(SLW BRK SOL)
			J1-T	24.0 ± 1 V dc
				(EL DR SOL)
SLEW	J1-FF	OPEN	J1-S	24.0 ± 1 V dc
				(TRK BRK SOL)
			J1-R	24.0 ± 1 V dc
				(SLW BRK SOL)
			J1-T	24.0 ± 1 V dc
				(EL DR SOL)

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INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
INTERRUPT	PANEL	NON-OVERRIDE	J1-s	0.5 ± 0.5 V d (AUD ALERT)

b. Weapon hatch interrupt/override. The following procedure shall be performed to verify that the laser interrupt and weapon hatch indicators illuminate, the laser cannot be fired and turret motion is inhibited when the weapon hatch has been opened. The stow position shall be allowed. It shall be verified that the override function enables firing of the laser and turret motion, and sounds the alarm. It shall be verified that turning off the override turns off the audible alarm. It shall be verified that closing the weapon hatch extinguishes the laser interrupt and weapon hatch indicators.

INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
WPN HTCH	J1-U	OPEN	J1-T	24.0 ± 1 V dc (EL DR SOL)
			J1-S	24.0 ± 1 V dc (TRK BRK SOL)
WPN HTCH	J1-U	0.5 ± 0.5 V dc (at 2 mA max)	J1-T	1.5 V dc (max) (EL DR SOL)
			J1-S	1.5 V dc (max) (TRK BRK SOL)
WPN HTCH	J1-U	OPEN		
TRIG RT	J1-GG	24.0 ± 1 V dc	J1-L	0.5 ± 0.5 V dc (TRIG IND)
			J2-e	5.0 ± 0.1 V dc (FIRE REMOTE)
			LAMPS	LASER INTRPT - illuminated
			LAMPS	WPN HTCH - illuminated
WPN HTCH	J1-U	0.5 ± 0.5 V dc		

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INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
TRIG RT	J1-GG	24.0 ± 1 V dc	J1-L	0.5 ± 0.5 V dc (TRIG IND)
			J2-e	0.5 ± 0.5 V dc ($\overline{\text{FIRE REMOTE}}$)
			LAMPS	LASER INTRPT - not illuminated
			LAMPS	WPN HTCH - not illuminated
TRIG RT	J1-GG	OPEN		

c. Driver's hatch interrupt/override. The following procedure shall be performed to verify that the laser interrupt and driver's hatch indicators illuminate, the laser cannot be fired, and turret action is inhibited when the driver's hatch has been opened. The stow position shall be allowed. It shall be verified that the override function enables firing of the laser, turret motion and sounds the audible alarm. It shall also be verified that closing the driver's hatch extinguishes the laser interrupt and driver's hatch indicators.

INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
DRVR HTCH	J1-n	OPEN	J1-T	24.0 ± 1 V dc ($\overline{\text{EL DR SOL}}$)
			J1-S	24.0 ± 1 V dc ($\overline{\text{TRK BRK SOL}}$)
INTERRUPT	PANEL	OVERRIDE	J1-T	1.5 V dc (max) ($\overline{\text{EL DR SOL}}$)
			J1-S	1.5 V dc (max) ($\overline{\text{TRK BRK SOL}}$)
			J1-s	24.0 ± 1 V dc (AUD ALRT)
ERCT CONF	J1-Z	OPEN	J1-T	24.0 ± 1 V dc ($\overline{\text{EL DRV SOL}}$)
			J1-S	24.0 ± 1 V dc ($\overline{\text{TRK BRK SOL}}$)
			J1-s	24.0 ± 1 V dc (AUD ALRT)

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INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
INTERRUPT	PANEL	NOT OVERRIDE	J1-s	0.5 ± 0.5 V dc (AUD ALRT)
DRVR HTCH	J1-n	0.5 ± 0.5 V dc	J1-T	1.5 V dc (max) ($\overline{\text{EL DRV SOL}}$)
			J1-S	1.5 V dc (max) ($\overline{\text{TRK BRK SOL}}$)
SLEW	J1-FF	24.0 ± 1 V dc	J1-S	24.0 ± 1 V dc ($\overline{\text{TRK BRK SOL}}$)
			J1-R	1.5 V dc (max) ($\overline{\text{SLW BRK SOL}}$)
			J1-T	1.5 V dc (max) ($\overline{\text{EL DR SOL}}$)

d. Cargo hatch interrupt/override. The following procedure shall be performed to verify that the laser interrupt and cargo hatch indicators illuminate and the laser cannot be fired when the cargo hatch has been opened (pop-up or full). When the cargo hatch is open (pop-up), it shall be verified that stow position and turret motion are also inhibited. It shall be verified that the override function enables firing of the laser, turret motion and sounds the audible alarm. It shall be verified that the stow position, with a popped-up cargo hatch cannot be overridden. It shall be verified that turning off the override turns off the audible alarm. It shall be verified that closing the cargo hatch extinguishes the laser interrupt and cargo hatch indicators.

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INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
CRGO HTCH	J1-p	OPEN	LAMPS -	CARGO HTCH - illuminated
DRIVE	PANEL	ON	J1-i	24.0 ± 1 V dc
				($\overline{\text{CCW DR SOL}}$)
DRIVE	PANEL	OFF		
CRGO HTCH FOPN	J1-g	0.5 ± 0.5 V dc		
DRIVE	PANEL	ON	J1-i	1.5 V dc (max)
				($\overline{\text{CCW DR SOL}}$)
DRIVE	PANEL	OFF	J1-i	24.0 ± 1 V dc
				($\overline{\text{CCW DR SOL}}$)
CRGO HTCH	J1-p	OPEN	LAMPS -	CRGO HTCH - illuminated
TRIG RT	J1-GG	24.0 ± 1 V dc	J2-e	5.0 ± 0.1 V dc
				($\overline{\text{FIRE REMOTE}}$)
			LAMPS -	LASER INTRPT - illuminated
CARGO HATCH	J1-p	0.5 ± 0.5 V dc	LAMPS	CRGO HTCH - not illuminated
			LAMPS	LASER INTRPT - not illuminated
TRIG RT	J1-GG	24.0 ± 1 V dc	J2-e	0.5 ± 0.5 V dc
				($\overline{\text{FIRE REMOTE}}$)
TRIG RT	J1-GG	OPEN	J2-e	5.0 ± 0.5 V dc
				($\overline{\text{FIRE REMOTE}}$)

e. WFOV interrupt. The following procedure shall be performed to verify that the laser interrupt indicator illuminates and the laser cannot be fired when the 3X optics has been selected. It shall be verified that the override function does not enable firing of the laser.

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INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
INTERRUPT	PANEL	OVERRIDE	LAMPS	LASER INTRPT - illuminated
TRIG RT	J1-GG	24.0 ± 1 V dc	J2-e	0.5 ± 0.5 V dc
				(FIRE REMOTE)
WFOVC	J1-BB	0.5 ± 0.5 V dc		
TRIG RT	J1-GG	24.0 ± 1 V dc	J2-e	5.0 ± 0.1 V dc
				(FIRE REMOTE)
WFOVC	J1-BB	OPEN		
TRIG RT	J1-GG	24.0 ± 1 V dc	J2-e	0.5 ± 0.5 V dc
				(FIRE REMOTE)

f. Shorted trigger switch interrupt. The following procedure shall be performed to verify that when one or both trigger switches on the hand control unit is depressed or failed during TSCD power-up and no fire interrupts exist with the G/VLLD power on, the laser cannot be fired. It shall be verified that a flashing "TRIG SHORT" operator error message appears on the TSCD display. It shall be verified that the override function does not enable firing of the laser.

INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
TSCD	PANEL	OFF		
TRIG RT	J1-GG	24.0 ± 1 V dc		
TSCD	PANEL	ON	J2-e	5.0 ± 0.1 V dc
				(FIRE REMOTE)
			DISPLAY	"TRIG SHORT"
			LAMPS	LASER INTRPT - illuminated
INTERRUPT	PANEL	OVERRIDE	J2-e	5.0 ± 0.1 V DC
				(FIRE REMOTE)
INTERRUPT	PANEL	NOT OVERRIDE		
TRIG RT	J1-GG	OPEN	LAMPS	LASER INTRPT - not illuminate

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g. Stow interrupts. The following procedure shall be performed to verify that the targeting head cannot be commanded to the stow position unless both the EL STOW and AZ STOW indicators not illuminated, indicating that the proper stow position has been achieved. The override function shall not allow the head to be stowed.

INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
EL STOW	J1-V	OPEN		
AZ STOW	J1-r	24.0 ± 1 V dc	LAMPS	AZ STOW - illuminated
HEAD	PANEL	STOW		
DRIVE	PANEL	ON	J1-i	24.0 ± 1 V dc ($\overline{\text{CCW DR SOL}}$)
DRIVE	PANEL	OFF	J1-i	24.0 ± 1 V dc ($\overline{\text{CCW DR SOL}}$)
EL STOW	J1-V	24.0 ± 1 V dc	LAMPS	EL STOW - illuminated
DRIVE	PANEL	ON	J1-i	1.5 V dc (max) ($\overline{\text{CCW DR SOL}}$)
DRIVE	PANEL	OFF	J1-i	24.0 ± 1 V dc ($\overline{\text{CCW DR SOL}}$)
AZ STOW	J1-r	OPEN	LAMPS	AZ STOW - not illuminated
DRIVE	PANEL	ON	J1-i	24.0 ± 1 V dc ($\overline{\text{CCW DR SOL}}$)
DRIVE	PANEL	OFF	J1-i	24.0 ± 1 V dc ($\overline{\text{CCW DR SOL}}$)

h. G/VLLD mount interrupt. The following procedure shall be performed to verify that turret motion is not allowed when the G/VLLD mount is unlocked (travel mode). It shall be verified that the override function sounds the alarm and allows turret motion.

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INPUT SIGNALS			OUTPUT SIGNALS	
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
STOW CONF	J1-Y	OPEN		
GLLD LKS	J1-e	OPEN		
ERCT LKS	J1-b	24.0 ± 1 V dc	LAMPS	HEAD ERCT illuminated
SLEW	J1-FF	24.0 ± 1 V dc	J1-S	24.0 ± 1 V dc (TRK BRK SOL)
			J1-R	24.0 ± 1 V dc (SLW BRK SOL)
			J1-T	24.0 ± 1 V dc (EL DR SOL)
SLEW INTERRUPT	J1-FF PANEL	OPEN OVERRIDE	J1-s	24.0 ± 1 V dc (AUD ALERT)
			J1-S	1.5 V dc (max (TRK BRK SOL)
			J1-T	1.5 V dc (max (EL DR SOL)
SLEW	J1-FF	24.0 ± 1 V dc	J1-S	24.0 ± 1 V dc (TRK BRK SOL)
			J1-R	1.5 V dc (max (SLW BRK SOL
			J1-T	1.5 V dc (max (EL DR SOL)
SLEW INTERRUPT	J1-FF PANEL	OPEN NON-OVERRIDE	J1-s	0.5 ± 0.5 V dc (AUD ALRT)
GLLD LKS	J1-c	24.0 ± 1 V dc	LAMPS	HEAD READY - illuminated

i. Erect lock interrupt. The following procedure shall be performed to verify that turret motion is not allowed when the erection locks are not engaged. It shall be verified that the override function sounds the alarm and allows turret motion.

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INPUT SIGNALS			OUTPUT	
SIGNALS				
SIGNAL NAME	LOCATION	INPUT CONDITION	LOCATION	SIGNAL LEVEL
STOW CONF	J1-Y	OPEN		
GLLD LKS	J1-e	24.0 ± 1 V dc		
ERCT LKS	J1-b	OPEN		
SLEW	J1-FF	24.0 ± 1 V dc	J1-S	24.0 ± 1 V dc
				$\overline{\text{(TRK BRK SOL)}}$
			J1-R	24.0 ± 1 V dc
				$\overline{\text{(SLW BRK SOL)}}$
			J1-T	24.0 ± 1 V dc (EL $\overline{\text{DR}}$ SOL)
SLEW	J1-FF	OPEN		
INTERRUPT	PANEL	OVERRIDE	J1-s	24.0 ± 1 Vdc (AUD $\overline{\text{ALRT}}$)
			J1-S	1.5 V dc (max)
				$\overline{\text{(TRK BRK SOL)}}$
			J1-T	1.5 V dc (max)
				$\overline{\text{(EL DR SOL)}}$
SLEW	J1-FF	24.0 ± 1 V dc	J1-S	24.0 ± 1 V dc
				$\overline{\text{(TRK BRK SOL)}}$
			J1-R	1.5 V dc (max)
				$\overline{\text{(SLW BRK SOL)}}$
			J1-T	1.5 V dc (max)
				$\overline{\text{(EL DR SOL)}}$
SLEW	J1-FF	OPEN		
INTERRUPT	PANEL	NON-OVERRIDE	J1-s	0.5 ± 0.5 V dc (AUD $\overline{\text{ALRT}}$)
ERCT LKS	J1-e	24.0 ± 1 V dc	LAMPS	HEAD ERECT - illuminated HEAD READY - illuminated

40.1.19 Voltage test. To verify conformance to 3.5.2, the TSCD shall be supplied with 24 ± 6 V dc and shall operate as specified herein. The voltage (applied to pin J1-w) shall be decreased to +20.0 V dc and verified that the "EQPT BAT" low voltage lamp illuminates. Pin J1-z shall be supplied with +24 V dc and it shall be verified that the "VEH BAT" low voltage lamp illuminates when this voltage is lowered to +16.0 V dc.

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